# Alt–Zeta: A New Compass on the Road to Riemann

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**Abstract**

*Aim.* The Riemann Hypothesis (RH) is the organizing conjecture of analytic number theory. We pursue RH through a *finite*, dyadic, and verifiable surrogate: an *alt–zeta* whose completion satisfies an exact functional equation, whose explicit formula is band-limited with nonnegative zero-weights, and whose associated kernel is positive semidefinite (PSD). The strategy is to force a contradiction from any off-line zero using a *positivity barrier* on each dyadic window.

*Strategy: why an alt–zeta?* The classical $\zeta$ simultaneously entangles functional equation, Euler product, and explicit formula. In a finite regime one cannot keep all three without loss. We therefore engineer an *alternative* object with the two structural features that power positivity: (i) an exact functional equation by construction (odd completed log-derivative), and (ii) a smoothed explicit formula whose zero-term enters with a global *minus* sign and *nonnegative* weights (Fejér log-band). The arithmetic information is isolated in a contract, ETI (Extended Triple Interface), that supplies a dyadic pin and dispersion bounds. This yields a finite, machine-certifiable *zero-free horizontal band* for the completed object.

*Methodological aims.* (1) Work *dyadically* with a finite Dirichlet polynomial $B^{\text{fin}}$ and a completed $\Xi^{\text{fin}}$; (2) fix canonical kernels: Gaussian Mellin completion and Fejér band kernel on the log-line; (3) separate symmetry/positivity (engineered) from arithmetic inputs (ETI); (4) insist on *short-shift averages* only, avoiding single-shift dependence (e.g. $\Delta = 2$); (5) export deterministic JSON certificates (FE, PSD, margin, frame) with hashes for reproduction; (6) make the construction robust under small mask/kernel perturbations.

*Result (finite barrier).* On each dyadic $[X, 2X]$, under ETI with margin

$$M_H = c_0 - C_1\varepsilon_H - C_2H^{-1} - C_3e^{-B} > 0,$$

the completed finite object $\Xi^{\text{fin}}$ has *no zero off the critical line* within the effective band $|\Im s| \le 2c/H$. We also show test-function agreement between $-(\mathcal{Z}^{\text{fin}})'/\mathcal{Z}^{\text{fin}}$ and $-\zeta'/\zeta$ on $\Re s > 1$ at rate $O(H^{-1} + \varepsilon_H + e^{-B})$.

*Outlook.* We do not claim unconditional RH. A uniform ETI across heights would globalize the band-wise barrier. The design is *modular*: future modules (averaged Goldbach pins, twin-prime tilts) can strengthen ETI without altering the core machinery.

## Contents

# 1 Orientation and Roadmap

Our objective is to engineer a finite, dyadic-level object $\Xi^{\mathrm{fin}}$ with an exact functional equation and a band-limited explicit formula whose zero-weights are nonnegative and whose zero-term enters with a minus sign. This enables a positivity argument: any off-line zero would force a negative value of a PSD quadratic form, a contradiction provided the ETI pin exceeds the error ledger.

**Pipeline.** Fix $[X, 2X]$ and scales $H = (\log X)^A$, $Q = H^\gamma$, $N = 2Xe^{B/H}$. We specify: (i) Gaussian Mellin completion $\mathcal{M}J_H$, (ii) Fejér log-band kernel $K_H$, (iii) a $\delta = 2$ bank/mask, (iv) finite coefficients $b_{H,Q}$, (v) the completed finite object $\Xi^{\mathrm{fin}}$, (vi) the finite explicit formula with nonnegative weights, (vii) a PSD kernel and band coercivity, (viii) the positivity barrier. Certificates guarantee FE, PSD, margin, and sampling conditions.

## 1.1 What we mean by ETI

*Extended Triple Interface (ETI)* is the arithmetic middleware exported by our Goldbach/twin "triple-interface" stack. It packages only the short-shift, band-limited information that the analytic alt-zeta skeleton consumes. Concretely, ETI collects: (i) a *banked TFA layer* (Fejér-type bank on rationals $a/q$, $q \le Q$, width $\asymp 1/H$) with mean-zero projectors that kill constant/alias modes; (ii) *AO dispersion* (no small subfamily of arcs can capture more than its proportional bank energy); (iii) *BG/SSU* square-root short-shift savings in Type–II ranges; and (iv) a *Type–I ledger* (leakage $\ll X/(HQ^2)$ after banking). These blocks are documented for Goldbach and cloned for twins; the twin programme adds two twin-specific tests: *one-level spectral-null saving* and *cross-level decay*. ETI is precisely the finite set of constants and inequalities from these items that the alt-zeta pipeline imports; nothing else from Goldbach/twins is used here.

## 1.2 Main finite-conditional statement (diagnostic scope)

We fix the architecture of §3–4 and the ETI contract of §5.1 on a dyadic $[X, 2X]$. Write

$$M_H \overset{(57)}{=} c_0 - C_1\varepsilon_H - C_2H^{-1} - C_3e^{-B}, \qquad \mathcal{B}_H = \left\{\tfrac{1}{2} + i\xi : |\xi| \le 2c/H\right\}.$$

**Theorem 1.1** (Finite-conditional zero-free band)**.** *Assume* ETI$(X; H, Q)$ *(Assumption 5.2) with constants* $(c_0, \varepsilon_H, C_{AO,SSU}, m_H)$*, and build* $\Xi_{\rm fin}$ *as in §4. If for some* $x \in [X, 2X]$

$$M_H \;>\; C_\Gamma \;+\; x^{1/2}\,\mathsf{S}_H^{\rm cert},$$

*where* $C_\Gamma$ *bounds the completion term of* (47) *and* $\mathsf{S}_H^{\rm cert}$ *is a (prospective) upper bound for the spectral envelope on* $\mathcal{B}_H$ *(§11), then* $\Xi_{\rm fin}$ *has* no zeros off the critical line *in* $\mathcal{B}_H$*.*

This is a direct restatement of Theorem 9.1 specialized to the diagnostic inputs and the ledger $M_H$ of (57). The result is *finite-conditional*: it depends only on ETI, the schedule, and band-limited numerics (prospective certificates), not on any single-shift assertions.

**Corollary 1.1** (Carrier scan)**.** *Let* $K_{H,\tau}(z) = e^{i\tau z} K_H(z)$ *so* $\widehat{K_{H,\tau}}(t) = \widehat{K}_H(t-\tau)$*. If the hypotheses of Theorem 1.1 hold uniformly in* $\tau$ *on a compact height set, then the corresponding translated bands* $\{\,\frac{1}{2} + i\xi : |\xi - \tau| \le 2c/H\,\}$ *are zero-free off the critical line for* $\Xi_{\rm fin}$*.*

### 1.3 Editorial Warning on LLM Usage and Verification

What follows is a research plan. This manuscript was largely drafted with help from an LLM (MathGPT by Pulsr) that uses a Python backend to check for mathematical accuracy. Because LLMs can produce superficially plausible but ultimately incorrect arguments, all claims here should be treated as tentative until human verification. Acute skepticism is warranted and encouraged.

The human editor acted not as a traditional author who originates and certifies all arguments, but as an editor-curator – setting goals, posing falsifiers, choosing among LLM proposals, and cutting defective parts – without personally verifying every derivation. They are therefore listed as witness/editor, not as full author, because an LLM cannot bear scholarly responsibility and the editor will not claim it on its behalf.

Anyone citing this work should note that an LLM was used and that the human role was editorial. Final responsibility for correctness lies with the mathematical community's later process of verification, and any errors should be publicly recorded so future versions can be fixed. Any genuine mathematical contribution is ultimately due to the community whose work trained the model, and therein lies the credit. All copyright is released to the public domain. Finally, all discredit lies with the editor for wasting time and resources on what may turn out to be mathematical AI slop. Caveat emptor.

## 2 Preliminaries and Notation

This section fixes global conventions, symbols, and analytic transforms used throughout. All implicit constants in $O(\cdot)$ or $\ll$ may depend on the fixed schedule parameters $(A, \gamma, c, B)$ but are otherwise absolute.

### 2.1 Dyadic window and scale parameters

We work on a dyadic interval $[X, 2X]$ with

$$H = (\log X)^A, \qquad Q = H^\gamma \quad (0 < \gamma < \tfrac{1}{2}), \qquad N = 2X\, e^{B/H}, \quad B \ge B_0 > 0. \tag{1}$$

Here $H$ is the *bandwidth scale* on the log-line, $Q$ is the (minor/major-arc) auxiliary scale, and $N$ is the finite Dirichlet cut-off. We set $0 < c \le 1$ for the Fejér band parameter. Typical safe choices used later are $(A, \gamma, c, B) = (10, 0.3, \frac{1}{2}, 12)$.

## 2.2 Arithmetic functions and Dirichlet polynomials

We denote by $\Lambda$ the von Mangoldt function, by $\mu$ the Möbius function, and by $\Omega(n)$ the total number of prime factors of $n$ (with multiplicity). The *finite Dirichlet polynomials* we use are of the form

$$B^{\mathrm{fin}}(s) = \sum_{2\le n\le N} \frac{b_{H,Q}(n)}{n^s}, \tag{2}$$

with coefficients supported in $[2, N]$ and depending only on $(H, Q)$ and the fixed kernels; the concrete choice is specified in §4. We write $s = \sigma + it$ for the complex variable in the $s$-plane and use the shorthand

$$\frac{B^{\mathrm{fin}\prime}}{B^{\mathrm{fin}}}(s) = \frac{d}{ds}\log B^{\mathrm{fin}}(s) \qquad \text{whenever } B^{\mathrm{fin}}(s) \neq 0.$$

## 2.3 Log-line, Mellin/Fourier conventions

Let $z = \log x$ for $x > 0$. For functions $\phi : \mathbb{R} \to \mathbb{C}$ on the log-line we use the Fourier transform

$$\widehat{\phi}(t) = \int_{\mathbb{R}} \phi(z)\, e^{-itz}\, dz, \qquad \phi(z) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{\phi}(t)\, e^{itz}\, dt, \tag{3}$$

so Plancherel reads $\|\phi\|_2^2 = (2\pi)^{-1}\|\widehat{\phi}\|_2^2$. The Mellin transform is simply the Laplace transform in the $z$-variable,

$$\mathcal{M}\phi(s) = \int_{\mathbb{R}} \phi(z)\, e^{sz}\, dz, \qquad s \in \mathbb{C}, \tag{4}$$

and is related to (3) by $\mathcal{M}\phi(\frac{1}{2}+it) = \widehat{\phi}(t)\, e^{z/2}$ for the standard identification of $x = e^z$. In particular, when we speak of *bandwidth* we refer to the essential support of $\widehat{\phi}$ in the $t$-variable.

## 2.4 Smoothing on the log-line

Given a kernel $K : \mathbb{R} \to \mathbb{R}$ and an arithmetic function $a : \mathbb{N} \to \mathbb{C}$, we write

$$(a * K)(\log x) \stackrel{\mathrm{def}}{=} \sum_{n\ge 1} a(n)\, K(\log n - \log x). \tag{5}$$

The Fejér kernel $K = K_H$ used later is even, nonnegative, and band-limited:

$$\widehat{K_H}(t) = \max\Big(1 - \frac{H\,|t|}{2c}, 0\Big) \quad \text{with support} \quad |t| \le \frac{2c}{H}, \tag{6}$$

so that $K_H$ is real, nonnegative, and $\int_{\mathbb{R}} K_H(z)\, dz = 1$. The Mellin-even Gaussian completion kernel $\mathcal{M}J_H$ is entire, nonvanishing, and satisfies $\mathcal{M}J_H(1-s) = \mathcal{M}J_H(s)$. Their explicit formulas are recorded in §3.

## 2.5 Bandlimited test functions and sampling

We say $\phi$ is $1/H$*-bandlimited* if $\widehat{\phi}$ is supported in $|t| \le 2c/H$. In that case, classical Paley–Wiener theory and Kadec's 1/4-theorem (implying stability under small jitter) give sampling bounds: if $\{t_k\}$ is a near-uniform grid with step $h \le \pi H/4$ and jitter $|\delta_k| \le \delta h$ for $\delta < 1/4$, then there exists a frame constant $C_{\mathrm{frame}} \asymp h(1-4\delta)$ such that

$$C_{\mathrm{frame}}\, \|\phi\|_2^2 \;\ll\; \sum_k |\phi(t_k)|^2 \;\ll\; C_{\mathrm{frame}}^{-1}\, \|\phi\|_2^2. \tag{7}$$

We will use (7) to lower-bound quadratic forms built from samples of $\Xi^{\mathrm{fin}}(\frac{1}{2} + i\xi)$ on the effective band, cf. §8.

### 2.6 Asymptotic notation and uniformity

For functions $F = F(X, H, Q, \ldots)$ we write $F \ll G$ or $F = O(G)$ if $|F| \le C\,G$ with a constant $C$ depending only on $(A, \gamma, c, B)$ and fixed absolute constants. A statement holds *uniformly for* $x \in [X, 2X]$ if the implied constants do not depend on $x$ in that interval. When we write $o(1)$ it is with respect to $X \to \infty$ while keeping the schedule (1) (and thus $H, Q$) tied to $X$ as specified.

### 2.7 Error ledger shorthand

We frequently aggregate deterministic errors into the tripartite *ledger* bound

$$\mathcal{E}_H^{\rm fin} \;=\; O\Big(C_1\,\varepsilon_H \;+\; C_2\,H^{-1} \;+\; C_3\,e^{-B}\Big), \qquad \varepsilon_H \stackrel{\rm def}{=} \Big(\frac{H}{X}\Big)^{1/2} + \frac{1}{HQ^2}, \tag{8}$$

where $C_1, C_2, C_3$ are absolute constants attached to the fixed kernels and the sampling rule. The term with $\varepsilon_H$ is the ETI variance component; $H^{-1}$ is the kernel bandwidth tail; $e^{-B}$ is the finite cut-off remainder.

### 2.8 Objects of study (pointer)

The concrete kernels $(J_H, K_H)$ and coefficients $b_{H,Q}(n)$ are specified in §3 and §4, respectively. The completed finite object $\Xi^{\rm fin}$ is defined in §4 and satisfies an exact finite functional equation (Theorem 4.1). Its smoothed explicit formula with nonnegative zero-weights is stated in Theorem 7.1. The PSD kernel on the band and the coercivity bound appear in §8, and the finite positivity barrier is proved in §9.

## 3 Canonical Kernels and Bank

This section specifies the two canonical kernels used throughout and the discrete bank/mask infrastructure that interfaces with ETI. We also record elementary properties needed later.

### 3.1 Mellin-even Gaussian completion

Fix $\alpha_H = H^{-1}$. We *define* the Mellin transform of the completion kernel by

$$\mathcal{M}J_H(s) = \exp\Big(-\frac{(s-\frac12)^2}{2\alpha_H^2}\Big), \qquad s \in \mathbb{C}. \tag{9}$$

This choice is entire, nonvanishing, and *even about* $1/2$: $\mathcal{M}J_H(1-s) = \mathcal{M}J_H(s)$. We do not require a closed $z$-space formula for $J_H$; all uses of $J_H$ in the paper occur through its Mellin transform (9) or through smoothed coefficients $\Lambda_H = \Lambda * J_H$ where the convolution is taken on the log-line. (The $z$-profile of $J_H$ is the inverse Mellin transform of (9).)

**Lemma 3.1** (Basic bounds for $\mathcal{M}J_H$)**.** *For $s = \sigma + it$,*

$$|\mathcal{M}J_H(s)| = \exp\Big(-\frac{(\sigma-\frac12)^2 - t^2}{2\alpha_H^2}\Big), \qquad \frac{\mathcal{M}J_H'(s)}{\mathcal{M}J_H(s)} = -\frac{s-\frac12}{\alpha_H^2}.$$

*In particular,* $(\log \mathcal{M}J_H)'(s)$ *is odd about* $1/2$ *and* $(\log \mathcal{M}J_H)''(s) = -\alpha_H^{-2}$.

*Remark* 3.1. The sign pattern in (9) is chosen to simplify the proof that the completed object $\Xi^{\rm fin}$ has an exact functional equation (§4) and to keep the completion entire and nonvanishing. Any entire, nonvanishing, even-about-1/2 choice would suffice for the structural arguments; (9) is the canonical Gaussian choice.

### 3.2 Fejér log-band kernel

Let $W = c/H$ with $0 < c \le 1$. Define

$$K_H(z) = \frac{1}{\pi W}\Big(\frac{\sin(Wz)}{z}\Big)^2, \qquad z \in \mathbb{R}. \tag{10}$$

Its Fourier transform on the log-line is the *triangle* (Fejér) multiplier

$$\widehat{K_H}(t) = \int_{\mathbb{R}} K_H(z)\, e^{-itz}\, dz = \max\Big(1 - \frac{|t|}{2W}, 0\Big) = \max\Big(1 - \frac{H\,|t|}{2c}, 0\Big). \tag{11}$$

Hence $\widehat{K_H} \ge 0$, even, with compact support $|t| \le 2W = 2c/H$, and $\widehat{K_H}(0) = 1$.

**Lemma 3.2** (Normalization and positivity)**.** *$K_H$ is real, even, nonnegative, and $\int_{\mathbb{R}} K_H(z)\, dz = 1$.*

*Proof sketch.* The function $B_W(z) \overset{\text{def}}{=} \frac{\sin(Wz)}{\pi z}$ has Fourier transform $\widehat{B_W}(t) = \mathbf{1}_{[-W,W]}(t)$. Then $K_H(z) = (\pi W)^{-1}\big(\sin(Wz)/z\big)^2 = \pi\,(B_W \star B_W)(z)$ up to a harmless constant; consequently $\widehat{K_H} = \widehat{B_W} * \widehat{B_W}$ is the triangle function in (11), which is nonnegative and compactly supported. Evaluating at $t = 0$ gives $\int_{\mathbb{R}} K_H(z)\, dz = \widehat{K_H}(0) = 1$. □

*Remark* 3.2 (Band-limitation). If $\phi$ is $1/H$-bandlimited (i.e. $\widehat{\phi}$ vanishes off $|t| \le 2c/H$), then $\phi \mapsto \phi * K_H$ is a projection onto the same band and preserves pointwise sign at $z = 0$ since $K_H \ge 0$ and $\int K_H = 1$. This underlies the *sign control* in the explicit formula of §7: zeros contribute with a global minus sign and nonnegative weights $\widehat{K_H}$.

### 3.3 Bank partition and $\delta = 2$ mask

Let $M \asymp Q^2$ and set $\mathcal{B} = \{0, 1, \dots, M-1\}$ with cyclic addition. We use a finite mask $v = (v_j)_{j\in\mathcal{B}}$ with the $\delta = 2$ *balancing* constraints

$$\sum_{j\in\mathcal{B}} v_j = 0, \qquad \sum_{j\in\mathcal{B}} j\, v_j = 0, \qquad \|v\|_2 = 1. \tag{12}$$

We denote by $\mathfrak{M}$ the orbit of $v$ under cyclic shifts. Given a bounded sequence $(a_j)_{j\in\mathcal{B}}$, the masked average around index $k \in \mathcal{B}$ is

$$\mathcal{A}_v[a](k) = \sum_{j\in\mathcal{B}} v_j\, a_{k+j}. \tag{13}$$

**Lemma 3.3** (Mean/linear drift annihilation)**.** *If $a_j = A + Bj + r_j$ with $A, B \in \mathbb{C}$ and $(r_j)$ arbitrary, then $\mathcal{A}_v[a](k) = \sum_j v_j r_{k+j}$ for all $k$; in particular, constant and linear drift components are annihilated by any $\delta = 2$ mask $v$ obeying (12).*

*Proof.* By (12), $\sum_j v_j = 0$ and $\sum_j j v_j = 0$, so the contributions of $A$ and $Bj$ vanish identically, leaving $\sum_j v_j r_{k+j}$. □

**Lemma 3.4** (Existence of $\delta = 2$ masks)**.** *For every $M \ge 3$ there exists a nonzero $v \in \mathbb{R}^M$ satisfying (12). Moreover one can choose $v$ supported on at most three consecutive indices when $M \ge 5$, and in general $v$ may be constructed by orthogonal projection of any nonzero vector onto $\{1, j\}^{\perp}_{j\in\mathcal{B}}$ followed by normalization.*

*Remark* 3.3 (Interface with ETI). The AO/SSU components of ETI operate on mask-balanced combinations as in (13), so that large-scale mean or linear drifts across the bank do not pollute the variance budget. The scale $M \asymp Q^2$ ensures that the bank resolution matches the minor/major-arc granularity used in the dispersion estimates.

### 3.4 Summary of kernel–bank roles

- $\mathcal{M}J_H$ is the Mellin-even, nonvanishing completion factor used to enforce $\Xi^{\rm fin}(1-s) = \Xi^{\rm fin}(s)$ by construction (see §4).
- $K_H$ produces a band-limited explicit formula whose zero contributions carry a global *minus* sign and *nonnegative* weights $\widehat{K}_H$ (see §7).
- The bank $\mathcal{B}$ and mask $v$ implement $\delta = 2$ balancing so that AO/SSU in ETI controls short-shift fluctuations without contamination from mean/linear drift.

## 4 Finite Dirichlet Object and Completion

We define the finite coefficients, the finite Dirichlet polynomial $B^{\rm fin}$, and the completed object $\Xi^{\rm fin}$ used throughout. The choices are canonical within our architecture and chosen to enforce an exact functional equation and sign control in the explicit formula.

### 4.1 Smoothing operators on the log-line

For an arithmetic function $a : \mathbb{N} \to \mathbb{C}$ and a kernel $K : \mathbb{R} \to \mathbb{R}$, define the log-line convolution

$$(a * K)(n) \ \stackrel{\rm def}{=} \ \sum_{m\geq 1} a(m)\, K\big(\log m - \log n\big), \tag{14}$$

whenever the series converges absolutely (this is guaranteed for the kernels we fix). We write $\Lambda_H \stackrel{\rm def}{=} \Lambda * J_H$ and, for an even, compactly-supported bump $\omega_H : \mathbb{Z} \to [0, 1]$ with $\sum_\Delta \omega_H(\Delta) = 1$ and $\operatorname{supp}(\omega_H) \subset [-H, H]$, we define the short-shift average

$$\Pi_H^{\rm avg}(n) \ = \ \sum_{|\Delta|<H} \omega_H(\Delta)\, \Lambda(n)\Lambda(n + \Delta). \tag{15}$$

### 4.2 Finite coefficients and finite Dirichlet polynomial

Fix a bounded calibration parameter $\eta_{H,Q} \in \mathbb{R}$. We set

$$b_{H,Q}(n) \ = \ \Lambda_H(n) \ + \ \eta_{H,Q}\, \frac{\Pi_H^{\rm avg}(n)}{\log n}, \qquad 2 \leq n \leq N, \tag{16}$$

and define the finite Dirichlet polynomial

$$B^{\rm fin}(s) \ = \ \sum_{2\leq n\leq N} \frac{b_{H,Q}(n)}{n^s}, \qquad s \in \mathbb{C}. \tag{17}$$

Since the sum (17) is finite, $B^{\rm fin}$ is an *entire* function of $s$. The pair term in (16) is normalized by $\log n$ to match units with $-\zeta'/\zeta$ on $\Re s > 1$ and to allow a clean Euler–Hadamard splitting (see §6).

**Lemma 4.1** (Basic bounds). *For $2 \leq n \leq N$ one has $b_{H,Q}(n) \ll \log n$. Moreover, $\sum_{n\leq N} |b_{H,Q}(n)|^2 \ll N(\log N)^2$ with constants depending only on $(A, \gamma, c, B)$ and the fixed shapes of $J_H, \omega_H$.*

*Proof sketch.* $\Lambda_H$ is a log-line average of $\Lambda$ against the rapidly decaying $J_H$; thus $\Lambda_H(n) \ll \log n$. For $\Pi_H^{\rm avg}$ we note that $\omega_H$ has finite support and that $|\Lambda(n)\Lambda(n + \Delta)| \leq (\log n)(\log(n + |\Delta|)) \ll (\log n)^2$ uniformly for $|\Delta| \leq H = o(X)$ on the dyadic, whence the claims. $\square$

### 4.3 Completion and exact functional equation

We choose the completion factor to be the Mellin-even Gaussian from §3:

$$\mathbf{G}_H(s) \ \stackrel{\text{def}}{=} \ \mathcal{M}J_H(s) \ = \ \exp\Big(-\frac{(s-\frac{1}{2})^2}{2\alpha_H^2}\Big), \qquad \alpha_H = H^{-1}. \tag{18}$$

This factor is entire, nonvanishing, and satisfies $\mathbf{G}_H(1-s) = \mathbf{G}_H(s)$. We then define the completed finite object

$$\Xi^{\text{fin}}(s) \ \stackrel{\text{def}}{=} \ \exp\Big(\int_{1/2}^{s} \frac{B^{\text{fin}}(u) - B^{\text{fin}}(1-u)}{2}\, du\Big)\ \mathbf{G}_H(s). \tag{19}$$

**Theorem 4.1** (Finite FE)**.** $\Xi^{\text{fin}}(1-s) = \Xi^{\text{fin}}(s)$ *for all* $s \in \mathbb{C}$.

*Proof.* Differentiate $\log \Xi^{\text{fin}}$:

$$(\log \Xi^{\text{fin}})'(s) = \frac{B^{\text{fin}}(s) - B^{\text{fin}}(1-s)}{2} + (\log \mathbf{G}_H)'(s).$$

The first term is odd under $s \mapsto 1-s$ by inspection, and $(\log \mathbf{G}_H)'$ is odd because $\mathbf{G}_H(1-s) = \mathbf{G}_H(s)$. Hence $(\log \Xi^{\text{fin}})'(1-s) = -(\log \Xi^{\text{fin}})'(s)$. Integrating along the segment from $1/2$ to $s$ and using $\Xi^{\text{fin}}(1/2) = \mathbf{G}_H(1/2)$ shows $\Xi^{\text{fin}}(1-s) = \Xi^{\text{fin}}(s)$. □

### 4.4 Smoothed prime functional and ETI interface

Define the smoothed prime functional (log-line convolution at scale $H$)

$$\Psi_{H,Q}(x) \ \stackrel{\text{def}}{=} \ \sum_{n\geq 2} b_{H,Q}(n)\, J_H(\log n - \log x), \qquad x \in [X, 2X]. \tag{20}$$

In the ETI contract (§5.1) we assume a *pin* $c_0 > 0$ for $\Psi_{H,Q}$, an $L^2(dx/x)$ variance budget $\varepsilon_H$, AO/SSU control after $\delta = 2$ masking across the bank, and a band nonvanishing constant $m_H$ for $\Xi^{\text{fin}}(\frac{1}{2} + i\xi)$ on $|\xi| \leq 2c/H$.

*Remark* 4.1 (Calibration parameter)*.* The scalar $\eta_{H,Q}$ in (16) may be chosen by a one-dimensional calibration (e.g. matching a main term or minimizing the $L^2(dx/x)$ deviation of $\Psi_{H,Q}$ from a target level). Our analysis treats $\eta_{H,Q}$ as fixed and bounded; any resulting change is absorbed into the ETI constants $c_0$ and $\varepsilon_H$.

### 4.5 Notes on growth and regularity

The function $B^{\text{fin}}$ is entire and of exponential type $\ll (\log N)$ because it is a finite Dirichlet polynomial; $\mathbf{G}_H$ is entire of Gaussian type in $t = \Im s$ with even symmetry about $1/2$. Consequently $\Xi^{\text{fin}}$ is entire and of finite (Gaussian) order, and $(\log \Xi^{\text{fin}})'$ is meromorphic with simple poles at the zeros of $\Xi^{\text{fin}}$ with residues equal to their multiplicities. These regularity properties suffice for the contour shifts used in the smoothed explicit formula (§7).

## 5 Extended Triple Interface (ETI) Contract

This section formalizes the arithmetic interface (ETI) that the analytic machinery will consume. All claims in the sequel that mention primes/pairs are routed *only* through ETI. The contract is stated for a fixed dyadic window $[X, 2X]$ and the fixed architecture $(H, Q, J_H, K_H, \mathcal{B}, v, \omega_H, \eta_{H,Q})$ specified earlier.

## 5.1 Extended Triple Interface (ETI): data, contract, and scope

**Purpose.** ETI formalizes the *only* arithmetic inputs our analytic machinery will consume. It is stated per dyadic $[X, 2X]$ and a fixed architecture $(H, Q; J_H, K_H; \text{bank } B,\ \delta = 2 \text{ mask } v;\ \omega_H;\ \eta_{H,Q})$. The finite Dirichlet object and completion use these choices, and all prime/pair content enters *solely* via ETI.

## 5.2 Data and derived functionals

Besides $(H, Q)$ and kernels $(J_H, K_H)$, we fix:

- a bank $B = \{0, \ldots, M-1\}$ with $M \asymp Q^2$ and a balanced $\delta = 2$ mask $v = (v_j)$; an even bump $\omega_H$ supported on $[-H, H]$ with $\sum_\Delta \omega_H(\Delta) = 1$; a bounded calibration scalar $\eta_{H,Q}$; :contentReference[oaicite:6]index=6
- the smoothed prime functional on the log-line

$$\Psi_{H,Q}(x) \ := \ \sum_{n \geq 2} b_{H,Q}(n)\, J_H(\log n - \log x), \qquad x \in [X, 2X],$$

  which interfaces ETI to the analytic side. :contentReference[oaicite:7]index=7

For masked bank dispersion, write $A_j(x)$ for the $j$-th bank-cell contribution to $\Psi_{H,Q}$ and define

$$D_v(x; k) \ := \ \sum_{j \in B} v_j\, A_{j+k}(x), \qquad D_v^*(x) \ := \ \sup_{k \in B} |D_v(x; k)|.$$

These are the AO/SSU-facing quantities that ETI will control. :contentReference[oaicite:8]index=8

## 5.3 The ETI contract

**Assumption 5.1** (ETI$(X; H, Q)$)**.** There exist constants $c_0 > 0$, $\varepsilon_H \geq 0$, $C_{\mathrm{AO,SSU}} \geq 1$, $m_H > 0$ (depending only on the fixed architecture and the dyadic) such that:

(E1) **PIN (lower floor).** For all $x \in [X, 2X]$, $\Psi_{H,Q}(x) \ \geq \ c_0$. :contentReference[oaicite:9]index=9

(E2) **Ledger (variance budget).** With the $L^2(dx/x)$ norm,

$$\int_X^{2X} \left|\Psi_{H,Q}(x) - c_0\right|^2 \frac{dx}{x} \ \leq \ \varepsilon_H^2\, X, \quad \varepsilon_H \ \asymp \ \left(\tfrac{H}{X}\right)^{1/2} + \tfrac{1}{HQ^2}.$$

:contentReference[oaicite:10]index=10

(E3) **AO/SSU (masked short-shift uniformity).**

$$\int_X^{2X} \left(D_v^*(x)\right)^2 \frac{dx}{x} \ \leq \ C_{\mathrm{AO,SSU}}\, \varepsilon_H^2\, X,$$

i.e. after $\delta = 2$ balancing no mean/linear drift across the bank can dominate the pair term. :contentReference[oaicite:11]index=11

(E4) **Band nonvanishing.** $\Xi_{\mathrm{fin}}(\frac{1}{2} + i\xi)$ has $|\Xi_{\mathrm{fin}}| \geq m_H$ on the effective band $|\xi| \leq 2c/H$ (used only for optional coercivity/envelope tightening). :contentReference[oaicite:12]index=12

**Scope and provenance.** Items (TFA/AO/BG/Type–I) arise from the triple-interface architecture introduced for Goldbach and ported to twins: Fejér-bank and alias suppression; deterministic AO dispersion; SSU square-root short-shift savings; and Type–I leakage $\ll X/(HQ^2)$ after banking. The twin stack adds *one-level spectral-null saving* and *cross-level decorrelation*, which are the only genuinely twin-specific novelties. ETI consumes only the finite consequences summarized above.

### 5.4 Goldbach-only mode ($\mathbf{ETI}_{\mathrm{GB}}$)

When only the Goldbach triple-interface is available, replace the pair input by the Goldbach-only short-shift functional and define the coefficients $b^{\mathrm{GB}}_{H,Q}(n)$ accordingly. All results in this draft (explicit formula with nonnegative weights, PSD kernel, positivity barrier, schedule/rollout) hold verbatim under $\mathrm{ETI}_{\mathrm{GB}}$. :contentReference[oaicite:16]index=16

**Reader map.** For a one-page architectural view of TFA/AO/BG/Type–I and the twin-specific inserts, see the twin analytic outline; for the Goldbach bank/ledger scales and SSU savings at polylog ranges, see the Goldbach abstract and introduction.

### 5.5 Data and derived functionals

Besides $(H,Q)$ and the kernels $J_H, K_H$, we fix:

- a bank $\mathcal{B}=\{0,1,\dots,M-1\}$ with $M \asymp Q^2$ and a $\delta=2$ mask $v=(v_j)$ obeying (12);
- an even, nonnegative, compactly supported bump $\omega_H:\mathbb{Z}\to[0,1]$ with $\mathrm{supp}(\omega_H)\subset[-H,H]$ and $\sum_\Delta \omega_H(\Delta)=1$;
- the calibration scalar $\eta_{H,Q}\in\mathbb{R}$ (bounded, fixed once per dyadic).

Write the (log-line) smoothed prime functional

$$\Psi_{H,Q}(x)=\sum_{n\ge 2} b_{H,Q}(n)\,J_H(\log n-\log x),\qquad x\in[X,2X]. \tag{21}$$

For masked bank dispersion of pair terms, let $A_j(x)$ denote the contribution to $\Psi_{H,Q}(x)$ coming from the $j$-th bank cell (a standard partition of the major/minor-arc range compatible with $M$). Define the masked fluctuation at location $k\in\mathcal{B}$ by

$$\mathcal{D}_v(x;k)\;=\;\sum_{j\in\mathcal{B}} v_j\,A_{k+j}(x), \tag{22}$$

and the worst-case masked dispersion by $\mathcal{D}^*_v(x)=\sup_{k\in\mathcal{B}}|\mathcal{D}_v(x;k)|$.

### 5.6 The ETI contract

**Assumption 5.2** (ETI$(X;H,Q)$)**.** There exist constants

$$c_0>0,\qquad \varepsilon_H\ge 0,\qquad C_{AO,SSU}\ge 1,\qquad m_H>0,$$

depending only on the fixed architecture and on the dyadic $[X,2X]$, such that:

(E1) **PIN (lower floor).** For all $x\in[X,2X]$,

$$\Psi_{H,Q}(x)\;\ge\;c_0. \tag{23}$$

(E2) **Ledger (variance budget).** One has the $L^2(dx/x)$ control

$$\int_X^{2X} \left|\Psi_{H,Q}(x) - c_0\right|^2 \frac{dx}{x} \;\le\; \varepsilon_H^2\, X, \tag{24}$$

with the canonical scale

$$\varepsilon_H \;=\; \Big(\frac{H}{X}\Big)^{1/2} \;+\; \frac{1}{H\,Q^2}. \tag{25}$$

(E3) **AO/SSU (masked short-shift uniformity).** For the masked dispersion defined in (22), one has

$$\int_X^{2X} \big(\mathcal{D}_v^*(x)\big)^2 \frac{dx}{x} \;\le\; C_{AO,SSU}\, \varepsilon_H^2\, X. \tag{26}$$

In particular, after $\delta = 2$ balancing no large mean/linear drift across the bank can dominate the pair contribution.

(E4) **Band nonvanishing.** For the completed finite object $\Xi^{\mathrm{fin}}$ one has

$$\inf_{|\xi|\le 2c/H} \left|\Xi^{\mathrm{fin}}(\tfrac{1}{2} + i\xi)\right| \;\ge\; m_H. \tag{27}$$

*Remark* 5.1 (Scope and inputs)*.* ETI makes *no* single-shift assertions (e.g. at $\Delta = 2$); all pair information is short-shift averaged via $\omega_H$ and bank-balanced via $v$. The constants $(c_0, \varepsilon_H, C_{AO,SSU}, m_H)$ are the *only* arithmetic inputs consumed by our analytic arguments.

## 5.7 Immediate consequences

We collect simple estimates used repeatedly in the sequel.

**Lemma 5.1** (Mean and $L^1$ control)**.** *Under* (24)*, one has*

$$\left|\frac{1}{\log 2}\int_X^{2X} \Psi_{H,Q}(x)\,\frac{dx}{x} - c_0\right| \;\le\; \varepsilon_H, \qquad \int_X^{2X} \left|\Psi_{H,Q}(x) - c_0\right| \frac{dx}{x} \;\ll\; \varepsilon_H. \tag{28}$$

*Proof.* Cauchy–Schwarz and $\int_X^{2X} \frac{dx}{x} = \log 2$ give the first bound; the second follows similarly. □

**Lemma 5.2** (Monotonicity under coarsening)**.** *If $H$ is replaced by $\tilde H \ge H$ and $Q$ by $\tilde Q \le Q\,(\tilde H/H)^{1/2}$, then the canonical scale* (25) *satisfies $\tilde\varepsilon_{\tilde H} \;\ll\; \varepsilon_H$.*

*Sketch.* Increasing $H$ decreases $H^{-1}$ and $(H/X)^{1/2}$; decreasing $Q$ does not worsen $1/(HQ^2)$ beyond the displayed allowance. □

**Lemma 5.3** (Stability under small kernel/mask perturbations)**.** *Suppose* $\|\widehat{\delta K}\|_\infty \le \eta$*,* $\operatorname{supp}(\widehat{\delta K}) \subseteq \operatorname{supp}(\widehat{K_H})$*, and $v$ is replaced by a cyclic shift $\tilde v \in \mathfrak{M}$. Then the ETI constants change by at most $O(\eta)$ multiplicative factors. In particular, if* (23)*–*(27) *hold for $(K_H, v)$, they hold with proportionally perturbed constants for $(K_H + \delta K, \tilde v)$.*

*Sketch.* All ETI quantities are continuous in the inputs by dominated convergence on $dx/x$ and by the bounded support of $\widehat{K_H}$; cyclic shifts preserve the $\delta = 2$ constraints. □

### 5.8 Admissible choices and normalizations

A concrete admissible family for $\omega_H$ is the discrete Fejér bump

$$\omega_H(\Delta) = \frac{1}{Z_H} \max\Big(1 - \frac{|\Delta|}{H}, 0\Big), \qquad Z_H = \sum_{|\Delta|<H} \max\Big(1 - \frac{|\Delta|}{H}, 0\Big), \tag{29}$$

which is even, nonnegative, supported in $[-H, H]$, and sums to 1. For the bank size we fix $M = \lfloor c_Q Q^2 \rfloor$ with a constant $c_Q \in [1, 4]$; results are insensitive to the choice of $c_Q$. The calibration $\eta_{H,Q}$ is chosen once per dyadic (e.g. by minimizing the $L^2(dx/x)$ deviation of $\Psi_{H,Q}$ from a target), and its effect is absorbed into $(c_0, \varepsilon_H)$.

### 5.9 Where ETI enters later

- In the smoothed explicit formula (§7), the ledger bound controls the error term and, together with the band limit of $\widehat{K}_H$, localizes zero contributions to $|\Im s| \le 2c/H$.
- In the PSD kernel (§8), the constant $m_H$ furnishes a coercivity bound on band-limited quadratic forms.
- In the positivity barrier (§9), the *margin* $M_H = c_0 - C_1\varepsilon_H - C_2 H^{-1} - C_3 e^{-B}$ combines ETI and analytic tails; if $M_H > 0$ there are no off-line zeros in the band.

### 5.10 Goldbach-only ETI ($\mathbf{ETI_{GB}}$)

In contexts where only the Goldbach triple-interface is available, we work with a restricted pair functional $\Pi_H^{\mathrm{GB}}(n)$ that encodes *short-shift* pair contributions obtained from Goldbach alone (no twin-prime input). Define the Goldbach-only coefficients

$$b_{H,Q}^{\mathrm{GB}}(n) \ := \ \Lambda_H(n) \ + \ \eta_{H,Q} \, \frac{\Pi_H^{\mathrm{GB}}(n)}{\log n}\,, \tag{30}$$

and the corresponding finite Dirichlet object $B_H^{\mathrm{fin,GB}}(s) = \sum_{2\le n\le N} b_{H,Q}^{\mathrm{GB}}(n)\, n^{-s}$ and completion $\Xi_{\mathrm{fin}}^{\mathrm{GB}}$ built exactly as in §4.

**Assumption 5.3** ($\mathrm{ETI_{GB}}(X; H, Q)$)**.** There exist $(c_0, \varepsilon_H, C_{AO,SSU}, m_H)$ such that the pin, ledger, AO/SSU, and band nonvanishing statements of Assumption 5.2 hold with $\Psi_{H,Q}$ formed from $b_{H,Q}^{\mathrm{GB}}$ in place of $b_{H,Q}$.

**Scope.** All analytic results in Draft 2 (smoothed explicit formula with sign/weights, PSD kernel, coercivity, positivity barrier, schedule, rollout) go through unchanged under Assumption 5.3, because only the short-shift, bank-balanced consequences of pairs are used. Twin-prime information is *not* required for any result stated in this draft.

## 6 Finite Euler–Hadamard Expansion on $\Re s > 1$

We record a decomposition of the finite Dirichlet object into prime-power (Euler) and short-shift pair (Hadamard) contributions, with an absolutely convergent remainder on $\Re s > 1$. This is the vehicle for later comparisons with $-\zeta'/\zeta$ and for bookkeeping the pair term introduced by $\Pi_H^{\mathrm{avg}}$.

## 6.1 Statement of the expansion

Recall $b_{H,Q}(n) = \Lambda_H(n) + \eta_{H,Q}\, \Pi_H^{\mathrm{avg}}(n)/\log n$ and $B^{\mathrm{fin}}(s) = \sum_{2\le n\le N} b_{H,Q}(n)\, n^{-s}$. Writing $s = \sigma + it$ with $\sigma > 1$ and expanding the coefficients against prime powers and distinct prime pairs yields:

**Proposition 6.1** (Finite Euler–Hadamard expansion)**.** *For $\sigma > 1$,*

$$\sum_{2\le n\le N} \frac{b_{H,Q}(n)}{\log n}\, n^{-s} \;=\; \sum_{p}\sum_{m\ge 1} \frac{\alpha_{p^m}}{m\, p^{ms}} \;+\; \sum_{p\ne q} \frac{\beta_{p,q}}{(pq)^s} \;+\; R^{\mathrm{fin}}(s), \tag{31}$$

*where*

$$\alpha_{p^m} \;:=\; \frac{\Lambda_H(p^m)}{\log(p^m)} \;=\; \frac{\Lambda_H(p^m)}{m\log p}, \qquad \beta_{p,q} \;:=\; \eta_{H,Q} \sum_{|\Delta|<H} \frac{\omega_H(\Delta)\, \mathbf{1}_{\,q-p=\Delta}}{\log p\, \log q}, \tag{32}$$

*and $R^{\mathrm{fin}}$ is an analytic function on $\{\sigma > 1\}$ with*

$$R^{\mathrm{fin}}(s) \;=\; O\big(H^{-1} \;+\; e^{-B}\big) \tag{33}$$

*uniformly on $\sigma \ge 1+\delta$ for each fixed $\delta > 0$. Both prime-power and pair sums are absolutely convergent on $\sigma > 1$.*

*Remark* 6.1. The normalization by $\log n$ in (31) is designed so that the Euler part matches the natural expansion of $-\zeta'/\zeta$ and the pair part has a clean $(pq)^{-s}$ structure with dimensionless coefficients $\beta_{p,q}$. The remainder $R^{\mathrm{fin}}$ aggregates the effects of finite cut-off $N$ and the mild mismatch between log-line smoothing and Dirichlet summation.

## 6.2 Proof of Proposition 6.1

We split the sum defining $B^{\mathrm{fin}}$ using $b_{H,Q} = \Lambda_H + \eta_{H,Q}\, \Pi_H^{\mathrm{avg}}/\log$. Since the sum is finite, rearrangements are harmless. For the prime-power part,

$$\sum_{2\le n\le N} \frac{\Lambda_H(n)}{\log n}\, n^{-s} \;=\; \sum_{p} \sum_{m\ge 1,\; p^m\le N} \frac{\Lambda_H(p^m)}{m\log p}\, p^{-ms}.$$

Extending $m$ to $\infty$ introduces an absolutely convergent tail (since $\sigma > 1$), bounded by $O\big(\sum_p \sum_{m>\log N/\log p} p^{-m\sigma}\big) = O(N^{-(\sigma-1)})$, which we absorb into $R^{\mathrm{fin}}$. This yields the first double sum in (31) with $\alpha_{p^m}$ as in (32).

For the pair part we expand

$$\sum_{2\le n\le N} \frac{\Pi_H^{\mathrm{avg}}(n)}{\log n}\, n^{-s} \;=\; \sum_{2\le n\le N} \sum_{|\Delta|<H} \omega_H(\Delta)\, \frac{\Lambda(n)\Lambda(n+\Delta)}{\log n}\, n^{-s}.$$

Writing $n = p$ and $n+\Delta = q$ when both are prime (the prime-power cases contribute to $R^{\mathrm{fin}}$ and are absolutely convergent on $\sigma > 1$), one obtains

$$\sum_{p\ne q} \sum_{|\Delta|<H} \omega_H(\Delta)\, \frac{\mathbf{1}_{\,q-p=\Delta}}{\log p}\, p^{-s} \cdot \frac{\Lambda(q)}{\log q}\, q^{-s} \;=\; \sum_{p\ne q} \frac{\beta_{p,q}}{(pq)^s} \;+\; R^{\mathrm{fin}}(s),$$

with $\beta_{p,q}$ as in (32). Absolute convergence on $\sigma > 1$ follows from $\sum_{p\ne q} (pq)^{-\sigma} < \infty$. Collecting the finite truncation and nonprime-power terms into $R^{\mathrm{fin}}$ gives (31).

It remains to justify (33). The cut-off $N = 2Xe^{B/H}$ pushes tails well beyond the dyadic; together with the rapid decay of $J_H$ this yields an $O(e^{-B})$ bound for differences between infinite and finite Euler sums on $\sigma > 1+\delta$. The log-line band limitation of $K_H$ and the short support of $\omega_H$ lead to $O(H^{-1})$ discrepancies in replacing discrete shifts with their band-truncated transforms; these are recorded in $R^{\mathrm{fin}}$. Details follow standard arguments in smoothed explicit-formula manipulations and are omitted.

### 6.3 Comparison with $-\zeta'/\zeta$ on $\Re s > 1$

Let

$$\mathcal{Z}^{\mathrm{fin}}(s) \overset{\mathrm{def}}{=} \sum_{2 \le n \le N} \frac{\Lambda_H(n)}{n^s}. \tag{34}$$

Formally, ignoring the finite cut-off and using the Mellin factorization,

$$\sum_{n \ge 2} \frac{\Lambda_H(n)}{n^s} \;=\; \mathcal{M}J_H(s) \sum_{n \ge 2} \frac{\Lambda(n)}{n^s} \;=\; \mathcal{M}J_H(s) \Big( -\frac{\zeta'}{\zeta}(s) \Big). \tag{35}$$

The finite cut-off error is bounded by $O(e^{-B})$ uniformly on $\sigma \ge 1 + \delta$. Hence we record:

**Lemma 6.1** (Finite-to-classical link)**.** *For each $\delta > 0$ and $\sigma \ge 1 + \delta$,*

$$\mathcal{Z}^{\mathrm{fin}}(s) \;=\; \mathcal{M}J_H(s) \Big( -\frac{\zeta'}{\zeta}(s) \Big) \;+\; O(e^{-B}). \tag{36}$$

*The implicit constant depends on $\delta$ and the fixed kernels only.*

*Remark* 6.2 (Test-function agreement)*.* Let $\psi \in C_c^\infty(\mathbb{R})$ with $\|\psi\|_{H^1}, \|\psi\|_{L^2} \le 1$. Then for $\sigma > 1$,

$$\int_{\mathbb{R}} \Big( B^{\mathrm{fin}} - \frac{\zeta'}{\zeta} \Big)(\sigma + it)\, \psi(t)\, dt \;=\; O\big(H^{-1} \;+\; \varepsilon_H \;+\; e^{-B}\big). \tag{37}$$

The terms correspond respectively to band tails, ETI ledger variance, and finite cut-off.

### Finite-conditional classical agreement (test–function level)

Let $\psi \in C_c^\infty(\mathbb{R})$ have $\|\psi\|_{H^1}, \|\psi\|_{L^2} \le 1$. Uniformly on $\sigma > 1$,

$$\int_{\mathbb{R}} \Big( B_{\mathrm{fin}} - \frac{\zeta'}{\zeta} \Big)(\sigma + it)\, \psi(t)\, dt \;=\; O\Big(H^{-1} + \varepsilon_H + e^{-B}\Big), \tag{38}$$

as in (68) and Remark **??**. Thus the finite object matches the classical logarithmic derivative *at test–function scale*, without invoking unproved global hypotheses.

### 6.4 Notes for later use

- The coefficients $\alpha_{p^m}$ and $\beta_{p,q}$ are nonnegative when $\eta_{H,Q} \ge 0$ and $\omega_H \ge 0$.
- The pair part is supported on $|q - p| < H$; this locality is crucial for AO/SSU control.
- Bounds such as $\sum_p \sum_{m \ge 1} \frac{\alpha_{p^m}}{m p^{m\sigma}} \ll 1$ for $\sigma > 1$ will be used when estimating residues in the smoothed explicit formula.

**GH-only mode.** All definitions and results hold with $b_{H,Q}$ replaced by $b_{H,Q}^{\mathrm{GB}}$ from (30). In particular, the Euler–Hadamard expansion and all band-limited explicit formulae remain valid: the only change is that the pair coefficients $\beta_{p,q}$ are computed from the Goldbach-only $\Pi_H^{\mathrm{GB}}$.

## 7 Smoothed Explicit Formula: Sign and Nonnegative Weights

We derive a smoothed explicit formula for the completed finite object $\Xi^{\mathrm{fin}}$ with a global *minus sign* in front of the zero sum and *nonnegative* spectral weights supported in the effective band. We also quantify the error ledger and bridge the ETI pin from the $J_H$-smoothed functional to the present $K_H$-smoothed one.

### 7.1 Band-limited smoothing of the finite coefficients

Define the $K_H$-smoothed finite prime functional

$$T_H(x) \stackrel{\text{def}}{=} \sum_{2\le n\le N} b_{H,Q}(n)\, K_H\big(\log n - \log x\big), \qquad x>0, \tag{39}$$

where $b_{H,Q}$ are the finite coefficients from §4. This choice aligns the prime side of the formula with the logarithmic derivative $(\log \Xi^{\text{fin}})'$, rather than with the classical $-\zeta'/\zeta$.

**Lemma 7.1** (Transfer from $J_H$ to $K_H$)**.** *Uniformly for $x\in[X,2X]$ one has*

$$T_H(x) \;=\; \Psi_{H,Q}(x) \;+\; O(H^{-1}), \tag{40}$$

*where $\Psi_{H,Q}$ is defined in (20). Consequently, the ETI pin and ledger transfer as*

$$T_H(x) \;\ge\; c_0 \;-\; C_2\, H^{-1} \;-\; C_1\, \varepsilon_H \quad \textit{for } x\in[X,2X] \textit{ in } L^2(dx/x)\textit{-sense.} \tag{41}$$

*Proof of Lemma 7.1.* Write the difference as

$$\Delta_H(x) \;:=\; T_H(x)-\Psi_{H,Q}(x) = \sum_{n\le N} b_{H,Q}(n)\,\big(K_H-J_H\big)(\log n-\log x),$$

where $b_{H,Q}$ are the finite coefficients from §4 and $K_H, J_H$ are the band kernels on the log-line (§3). Taking Mellin–Fourier transforms in the log variable, Plancherel on  gives

$$\int\!\big|\Delta_H(e^y)\big|^2\,dy = (2\pi)^{-1}\int\!\Big|K^\wedge_{c,H}(t)-J^\wedge_{c,H}(t)\Big|^2\Big|\sum_{n\le N} b_{H,Q}(n)\,n^{-it}\Big|^2\,dt,$$

where $K^\wedge_{c,H}$ and $J^\wedge_{c,H}$ are the compactly supported Fejér-type multipliers; by construction both are even, supported on $|t|\le 2c/H$, integrate to 1, and $\|K^\wedge_{c,H}-J^\wedge_{c,H}\|_{L^\infty()}\ll H^{-1}$. Hence

$$\int\!\big|\Delta_H(e^y)\big|^2\,dy \;\ll\; H^{-2}\int_{|t|\le 2c/H}\Big|\sum_{n\le N} b_{H,Q}(n)\,n^{-it}\Big|^2 dt.$$

By Cauchy–Schwarz and the $\ell^2$–bound for $b_{H,Q}$ (Lemma 4.1), $\sum_{n\le N}|b_{H,Q}(n)|^2\ll N(\log N)^2$, and since the $t$–integration spans length $\asymp H^{-1}$, we obtain

$$\int\!\big|\Delta_H(e^y)\big|^2\,dy \;\ll\; H^{-3}\,N(\log N)^2.$$

Fixing a dyadic $[X,2X]$ and using that $N\asymp Xe^{B/H}$ with $B\gg 1$ fixed at the dyadic, the $L^2(dy)$–bound implies the claimed $L^2(dx/x)$–bound $\|\Delta_H\|_{L^2([X,2X],\,dx/x)}\ll H^{-1}$. Finally, the ETI pin from §5 transfers from $\Psi_{H,Q}$ to $T_H$ by the triangle inequality, yielding the ledger statement in the lemma. ☐

### 7.2 The explicit formula

Let $\rho$ range over the (complex) zeros of $\Xi^{\text{fin}}$, counted with multiplicity. Recall that $\widehat{K_H}$, the Fourier transform of $K_H$, is even, nonnegative, and supported in $|t|\le 2c/H$.

**Theorem 7.1** (Smoothed explicit formula with band-limited weights)**.** *For $x > 0$,*

$$T_H(x) \;=\; \mathrm{Main}_H(x) \;-\; \sum_\rho \frac{x^\rho}{\rho}\,\widehat{K}_H\Big(\frac{\rho-\frac12}{i}\Big) \;+\; \mathcal{G}_H(x) \;+\; \mathcal{E}_H^{\mathrm{fin}}(x). \tag{42}$$

*Here $\widehat{K}_H \ge 0$ and is supported in $|\Im(\rho-\frac12)| \le 2c/H$, hence the zero sum is band-limited; $\mathcal{G}_H$ is a bounded gamma/completion contribution depending on $\mathbf{G}_H$; and the error ledger satisfies*

$$\|\mathcal{E}_H^{\mathrm{fin}}\|_{L^\infty([X,2X])} \;\ll\; C_1\,\varepsilon_H \;+\; C_2\,H^{-1} \;+\; C_3\,e^{-B}. \tag{43}$$

*Proof.* Consider the contour integral on $\Re s = 1+\varepsilon$,

$$I \;=\; \frac{1}{2\pi i}\int_{(1+\varepsilon)} \frac{\Xi^{\mathrm{fin}\prime}(s)}{\Xi^{\mathrm{fin}}(s)}\,\frac{x^s}{s}\,\widehat{K}_H\Big(\frac{s-\frac12}{i}\Big)\,ds, \tag{44}$$

where $\varepsilon > 0$ is fixed. The factor $\widehat{K}_H((s-\frac12)/i)$ is entire and rapidly decaying vertically, with compact support in the imaginary direction: $\widehat{K}_H((\sigma+it-\frac12)/i) = 0$ unless $|t| \le 2c/H$. Shift the contour to $\Re s = -\varepsilon$. Poles are at $s = \rho$ (zeros of $\Xi^{\mathrm{fin}}$) and at $s = 0$ (from the factor $1/s$). Residue calculation yields

$$I \;=\; -\sum_\rho \frac{x^\rho}{\rho}\,\widehat{K}_H\Big(\frac{\rho-\frac12}{i}\Big) \;+\; \mathrm{Res}_{s=0}\Big(\frac{\Xi^{\mathrm{fin}\prime}}{\Xi^{\mathrm{fin}}}(s)\frac{x^s}{s}\widehat{K}_H\Big(\frac{s-\frac12}{i}\Big)\Big) \;+\; E_{\mathrm{horiz}}, \tag{45}$$

where $E_{\mathrm{horiz}}$ is the sum of horizontal integrals; $E_{\mathrm{horiz}}$ vanishes because $\widehat{K}_H$ has compact support in $\Im(s)$ and $\Xi^{\mathrm{fin}}$ has at most Gaussian growth by construction (see §4).

Next, we evaluate $I$ by unfolding $(\log \Xi^{\mathrm{fin}})'(s) = \frac12(B^{\mathrm{fin}}(s) - B^{\mathrm{fin}}(1-s)) + (\log \mathbf{G}_H)'(s)$. Multiplying by $\frac{x^s}{s}\widehat{K}_H(\frac{s-\frac12}{i})$ and integrating on $\Re s = 1+\varepsilon$, the $B^{\mathrm{fin}}$-terms give

$$\frac{1}{2\pi i}\int_{(1+\varepsilon)} \Big(\tfrac12 B^{\mathrm{fin}}(s) - \tfrac12 B^{\mathrm{fin}}(1-s)\Big)\frac{x^s}{s}\widehat{K}_H\Big(\frac{s-\frac12}{i}\Big)\,ds \;=\; T_H(x) \;+\; O(e^{-B}), \tag{46}$$

where $T_H(x)$ appears from Mellin inversion and band-limitation, and $O(e^{-B})$ records the finite Dirichlet cut-off $N$. The completion term contributes

$$\mathcal{G}_H(x) \;\overset{\mathrm{def}}{=}\; \frac{1}{2\pi i}\int_{(1+\varepsilon)} (\log \mathbf{G}_H)'(s)\,\frac{x^s}{s}\,\widehat{K}_H\Big(\frac{s-\frac12}{i}\Big)\,ds, \tag{47}$$

which is bounded uniformly in $x$ because $(\log \mathbf{G}_H)'$ is entire and $\widehat{K}_H$ is compactly supported in $t$. Equating (45) with $I$ expressed as (46)+(47), and absorbing $O(e^{-B})$ into $\mathcal{E}_H^{\mathrm{fin}}$, yields (42). Finally, the ETI ledger and the transfer Lemma 7.1 contribute the remaining parts of (43). ☐

*Remark* 7.1 (Sign and nonnegative weights). The Fejér multiplier $\widehat{K}_H$ in (42) is *nonnegative* and even, so each zero contributes with weight $\widehat{K}_H((\rho-\frac12)/i) \ge 0$. The overall minus sign in front of the zero sum is a direct consequence of the residue computation in (45). Hence an off-line zero (with $\Re\rho \ne 1/2$) can be made to contribute a large negative *real* amount by aligning the phase of $x^\rho/\rho$ via the choice of $x$ in $[X, 2X]$ (see §9).

### 7.3 Gamma/completion and trivial terms

The function $\mathbf{G}_H$ is entire and even about 1/2, so $(\log \mathbf{G}_H)'$ is odd and analytic. The residue at $s = 0$ in (45) reduces to a bounded contribution (a constant depending only on fixed data) since $\widehat{K}_H((s - \frac{1}{2})/i)$ vanishes to first order at $s = 0$ by evenness and compact support. Consequently, $\mathcal{G}_H(x)$ in (42) is uniformly bounded in $x$ and contributes at most to the additive ledger.

### 7.4 Classical comparison (optional)

If one replaces $\Xi^{\text{fin}}$ with $\Xi$ built from the classical $\zeta$ (i.e. $(\log \Xi)'(s) = -\zeta'/\zeta(s) + (\log \Gamma\text{-part})'(s)$), then $T_H$ is replaced by $\sum \Lambda(n) K_H(\log n - \log x)$ and the same derivation yields the familiar explicit formula with the band-limited, nonnegative $\widehat{K}_H$. We emphasize that our positivity barrier relies on Theorem 7.1 for $\Xi^{\text{fin}}$ and the ETI transfer (41), not on classical single-shift inputs.

### 7.5 Carrier-shift invariance

[Carrier–shift invariance] For $\tau \in$ define $K_{H,\tau}(z) = e^{i\tau z} K_H(z)$ and

$$T_{H,\tau}(x) := \sum_{n \le N} b_{H,Q}(n)\, K_{H,\tau}(\log n - \log x)\,.$$

Then for every $x > 0$,

$$T_{H,\tau}(x) = x^{-i\tau} \sum_{n \le N} b_{H,Q}(n)\, n^{i\tau}\, K_H(\log n - \log x)\,.$$

In the explicit formula of Theorem 7.1, replacing $K_H$ by $K_{H,\tau}$ simply shifts the spectral weights $K_{c,H}$ to $K_{c,H}(\cdot - \tau)$ and introduces the phase $x^{-i\tau}$ on the main/gamma/error terms. In particular, the zero sum remains nonnegative–weighted and band–limited, with the band shifted by $\tau$.

*Proof.* The displayed identity is a direct algebraic rewrite using $K_{H,\tau}(z) = e^{i\tau z} K_H(z)$. Taking Fourier transforms in $z$ shows that $K^{\wedge}_{c,H,\tau}(t) = K^{\wedge}_{c,H}(t - \tau)$, so the weight applied to each zero $\rho$ in Theorem 7.1 becomes $K_{c,H}((\rho - \frac{1}{2})/i - \tau)$ while preserving nonnegativity and compact $t$–support. The factor $x^{-i\tau}$ pulls out of the contour integrals that define the main/gamma/error contributions, which leaves their absolute sizes unchanged and their phases aligned uniformly in $x$. $\square$

# 8 PSD Kernel on the Band and Coercivity

We build a de Branges–type positive semidefinite (PSD) kernel on the effective band $\mathcal{B}_H = [-2c/H, 2c/H]$ from the completed finite object $\Xi^{\text{fin}}$ and the Fejér band kernel. This kernel converts the minus/weight structure of the explicit formula (§7) into a quadratic form that must be nonnegative, thereby enabling the positivity barrier in §9.

### 8.1 Grids, weights, and atoms

Let $\{\xi_j\}_{j=1}^J \subset \mathcal{B}_H$ be sampling nodes on the band with positive weights $\{w_j\}_{j=1}^J$ (e.g. a near-uniform quadrature). Let $\{t_k\}_{k=1}^M \subset \mathcal{B}_H$ be "target" locations with mesh size at most $h$ and jitter at most $\delta h$ (see below). We use the *band-limited atoms*

$$\Phi_k(\xi) \overset{\text{def}}{=} \widehat{K}_H(\xi - t_k), \qquad \xi \in \mathbb{R}, \tag{48}$$

where $\widehat{K}_H$ is the Fejér multiplier from (11). Since $\widehat{K}_H$ has compact support in $\mathcal{B}_H$, each $\Phi_k$ is supported in the band.

## 8.2 Kernel construction and PSD

Define the $J \times M$ matrix $\mathbf{B}$ and diagonal $J \times J$ matrix $\mathbf{W}$ by

$$\mathbf{B}_{jk} \;=\; \frac{1}{\sqrt{2\pi}}\,\Xi^{\mathrm{fin}}\!\Big(\tfrac{1}{2}+i\xi_j\Big)\;\Phi_k(\xi_j), \qquad \mathbf{W} = \mathrm{diag}(w_1,\ldots,w_J), \tag{49}$$

and the $M \times M$ *band kernel matrix*

$$\mathbf{K} \;\stackrel{\mathrm{def}}{=}\; \mathbf{B}^{*}\,\mathbf{W}\,\mathbf{B}. \tag{50}$$

**Proposition 8.1** (PSD)**.** $\mathbf{K}$ *is Hermitian positive semidefinite. In fact, for any* $a \in \mathbb{C}^M$*,*

$$a^{*}\mathbf{K}\,a \;=\; \frac{1}{2\pi}\sum_{j=1}^{J} w_j \,\Big|\;\Xi^{\mathrm{fin}}\!\Big(\tfrac{1}{2}+i\xi_j\Big)\sum_{k=1}^{M} a_k\,\Phi_k(\xi_j)\;\Big|^2 \;\geq\; 0. \tag{51}$$

*Proof.* Immediate from (50) as a Gram matrix: $\mathbf{K} = \mathbf{B}^{*}\mathbf{W}\mathbf{B}$ with $\mathbf{W} \succeq 0$. □

*Remark* 8.1 (Continuous kernel)*.* In the quadrature limit, $\mathbf{K}$ discretizes the continuous PSD kernel

$$\mathsf{K}(t,u) \;=\; \frac{1}{2\pi}\int_{\mathcal{B}_H} \big|\Xi^{\mathrm{fin}}(\tfrac{1}{2}+i\xi)\big|^2\;\widehat{K}_H(\xi-t)\;\widehat{K}_H(\xi-u)\,d\xi, \tag{52}$$

which is evidently positive by Cauchy–Schwarz.

## 8.3 Sampling regime and frame bound

Assume the target grid $\{t_k\}$ is near-uniform with step $h$ and jitter $\delta h$:

$$t_k = \tau_0 + kh + \epsilon_k, \qquad |\epsilon_k| \le \delta h, \qquad 0 \le \delta < \tfrac{1}{4}, \tag{53}$$

and that $h \le \pi H/4$. Then the translates $\{\Phi_k\}$ form a (stable) frame for the Paley–Wiener band $\mathcal{B}_H$ in the sense that there exists $C_{\mathrm{frame}} > 0$ with

$$C_{\mathrm{frame}}\,\|f\|^2_{L^2(\mathbb{R})} \;\le\; \sum_{k=1}^{M} |(f,\Phi_k)_{L^2}|^2 \;\le\; C_{\mathrm{frame}}^{-1}\,\|f\|^2_{L^2(\mathbb{R})} \qquad \text{for all band-limited } f, \tag{54}$$

where $(\cdot,\cdot)$ is the $L^2(\mathbb{R})$ inner product.[1]

## 8.4 Coercivity under band nonvanishing

Assume ETI's *band nonvanishing* (27):

$$\inf_{|\xi|\le 2c/H} \big|\Xi^{\mathrm{fin}}(\tfrac{1}{2}+i\xi)\big| \;\geq\; m_H > 0.$$

Then $\mathbf{K}$ is not only PSD but *coercive* on the band-limited range of coefficients.

**Theorem 8.1** (Coercivity)**.** *Under* (53)*,* (54)*, and the band nonvanishing, there exists a constant*

$$\kappa_H \;=\; \frac{m_H^2}{2\pi}\,C_{\mathrm{frame}} \;> 0$$

*such that for all* $a \in \mathbb{C}^M$*,*

$$a^{*}\mathbf{K}\,a \;\geq\; \kappa_H\,\|a\|_2^2. \tag{55}$$

[1] For triangle $\widehat{K}_H$ this follows from standard shift-invariant frame theory for compactly supported band-limited generators combined with Kadec's 1/4-theorem for sampling stability; the constants depend continuously on $h$ and $\delta$.

*Proof sketch.* Let $F_a(\xi) = \sum_k a_k\, \Phi_k(\xi)$, which is supported in $\mathcal{B}_H$. By (51),

$$a^{*}\mathbf{K}\, a \;=\; \frac{1}{2\pi}\sum_j w_j \left|\Xi^{\mathrm{fin}}(\tfrac{1}{2}+i\xi_j)\right|^2 |F_a(\xi_j)|^2 \;\geq\; \frac{m_H^2}{2\pi}\sum_j w_j\, |F_a(\xi_j)|^2.$$

As the quadrature $\{(\xi_j, w_j)\}$ approximates the $L^2$ norm on $\mathcal{B}_H$, the RHS dominates $\frac{m_H^2}{2\pi}\|F_a\|_{L^2}^2$ up to a fixed constant that we absorb into $C_{\mathrm{frame}}$. Finally, the lower frame bound (54) implies $\|F_a\|_{L^2}^2 \geq C_{\mathrm{frame}}\, \|a\|_2^2$, giving (55). □

### 8.5 Numerical PSD certificate (discrete)

In computations we form the *scaled* kernel $\tilde{\mathbf{K}} = \mathbf{K}/\|\mathbf{K}\|_2$ and declare PSD certified if

$$\lambda_{\min}(\tilde{\mathbf{K}}) \;\geq\; \tau_{\mathrm{PSD}} \;\stackrel{\mathrm{def}}{=}\; 100\, M\, \varepsilon_{\mathrm{mach}}, \tag{56}$$

and coercivity certified if $\lambda_{\min}(\mathbf{K}) \geq \kappa_H^{\mathrm{num}}$, where $\kappa_H^{\mathrm{num}}$ is the computed lower bound for $(m_H^2/2\pi)\, C_{\mathrm{frame}}$ using the ETI value of $m_H$ and the sampling/frame parameters $(h, \delta)$.

### 8.6 How the kernel meets the explicit formula

Let $\varphi$ be the coefficient vector that encodes the band-limited test at $x \in [X, 2X]$ (via the Fejér smoothing and the grid $\{t_k\}$). The explicit formula in §7 expresses $T_H(x)$ as a main term minus a nonnegative-weighted zero sum, up to the ledger error. The kernel quadratic form $a^{*}\mathbf{K}a$ enforces nonnegativity for *every* such $a$. If an off-line zero existed in the band, we could choose $a = \varphi$ aligned so that the zero's contribution forces $a^{*}\mathbf{K}a$ below 0 unless the margin $M_H = c_0 - C_1\varepsilon_H - C_2 H^{-1} - C_3 e^{-B}$ is nonpositive. This contradiction proves the zero-free band once $M_H > 0$ (see §9).

*Remark* 8.2 (Non-circular usage of coercivity)*.* Coercivity (Theorem 8.1) is *not* invoked to prove zero-freeness. The barrier argument (Theorem 9.1) requires only that the Fejér weight $\widehat{K}_H$ be nonnegative and band-limited; the PSD property ensures quadratic nonnegativity for any test. Coercivity enters only as an *optional* tool to produce a numerically robust upper bound on the spectral envelope $\mathsf{S}_H$; a purely numeric, non-coercive fallback bound is provided in §15.11. Thus no circularity arises from assuming $m_H > 0$.

## 9 Finite Positivity Barrier (Zero-Free Band)

We now convert the minus/weight structure of the explicit formula (§7) and the PSD/coercivity of the band kernel (§8) into a *finite* zero-free statement on the effective band. Throughout this section we work on a fixed dyadic $[X, 2X]$ with the architecture and ETI constants of §5.1.

### 9.1 Margin and level calibration

Recall the *ledger* bound from (43) and Lemma 7.1. We aggregate deterministic tails into the *margin*

$$M_H \;\stackrel{\mathrm{def}}{=}\; c_0 \;-\; C_1\, \varepsilon_H \;-\; C_2\, H^{-1} \;-\; C_3\, e^{-B}. \tag{57}$$

By (41), $T_H(x) \geq M_H$ in the $L^2(dx/x)$ sense on $[X, 2X]$.

**Lemma 9.1** (Level calibration)**.** *There exists a bounded $\eta_{H,Q}$ (depending only on the dyadic and fixed kernels) such that*

$$\frac{1}{\log 2}\int_X^{2X} \mathrm{Main}_H(x)\,\frac{dx}{x} \;=\; c_0, \qquad \sup_{x\in[X,2X]} |\mathrm{Main}_H(x) - c_0| \;\ll\; H^{-1}.$$

*Proof.* Write $(\log \Xi_{\mathrm{fin}})' = \frac{1}{2}(B_H^{\mathrm{fin}} - B_H^{\mathrm{fin}}(1-\cdot)) + (\log G_H)'$. Testing against $\frac{x^s}{s}\widehat{K}_H(\frac{s-\frac{1}{2}}{i})$ and Mellin-inverting as in Theorem 7.1, the contribution of the symmetric Dirichlet part equals $T_H(x)$ and the gamma part integrates to a bounded $\mathcal{G}_H(x)$. Averaging over $[X,2X]$ with measure $dx/x$ amounts to evaluating at $s=1$ the Mellin transform of $K_H$ (which equals 1) and of the Fejér weight (which is even and band-limited); by (24), the average of $\Psi_{H,Q}(x)$ equals $c_0$ up to $O(\varepsilon_H)$, while Lemma 7.1 shows $T_H - \Psi_{H,Q} = O(H^{-1})$ in $L^2(dx/x)$. Choose $\eta_{H,Q}$ to make the average exact; boundedness of $\eta_{H,Q}$ follows from linearity and the normalization of $\omega_H$ and $K_H$. The pointwise $O(H^{-1})$ bound follows from band limitation and the Bernstein (Markov) inequality for $1/H$-band-limited functions on the log-line. □

## 9.2 Phase alignment on a dyadic

Write $\rho = \beta + i\gamma$. On a dyadic $[X,2X]$ we have $\log x \in [\log X, \log X + \log 2]$ and

$$\Re\Big(\frac{x^\rho}{\rho}\Big) = \frac{x^\beta}{|\rho|}\cos\big(\gamma\log x - \arg\rho\big).$$

Since $|\gamma| \le 2c/H$ in the band and $\log 2$ is fixed, the phase sweep on the dyadic has width at most $O(1/H)$. Therefore:

**Lemma 9.2** (Alignment)**.** *For every band zero $\rho$ there exists $x_\rho \in [X,2X]$ with*

$$\Re\Big(\frac{x_\rho^\rho}{\rho}\Big) \;\ge\; \kappa_{\mathrm{al}}\,\frac{x_\rho^\beta}{|\rho|}, \qquad \kappa_{\mathrm{al}} \;=\; \cos\Big(\frac{2c\,\log 2}{H}\Big) \;=\; 1 \;-\; O\Big(\frac{1}{H^2}\Big). \tag{58}$$

*Proof.* Let $\theta(x) = \gamma\log x - \arg\rho$; as $x$ ranges over a dyadic, $\theta$ traverses an interval of length at most $|\gamma|\log 2 \le (2c/H)\log 2$. Hence $\max_{x\in[X,2X]}\cos\theta(x) \ge \cos\big((2c/H)\log 2\big)$. □

## 9.3 A computable spectral envelope

Define the band-limited envelope

$$\mathsf{S}_H \;\stackrel{\mathrm{def}}{=}\; \sum_\rho \widehat{K}_H\Big(\frac{\rho-\frac{1}{2}}{i}\Big)\,\frac{1}{|\rho|}, \tag{59}$$

where the sum is restricted to the band $|\Im(\rho - \frac{1}{2})| \le 2c/H$. This quantity is finite (the weight has compact $t$-support) and *computable* from $\Xi^{\mathrm{fin}}$ via standard argument-principle quadrature; in practice we bound it by a *spectral envelope certificate* $\mathsf{S}_H^{\mathrm{cert}}$ (see §11).

**Lemma 9.3** (Rest-of-band bound)**.** *For any $x \in [X,2X]$,*

$$\sum_\rho \widehat{K}_H\Big(\frac{\rho-\frac{1}{2}}{i}\Big)\,\Big|\Re\Big(\frac{x^\rho}{\rho}\Big)\Big| \;\le\; x^{1/2}\,\mathsf{S}_H.$$

*Proof.* $|\Re(x^\rho/\rho)| \le |x^\rho|/|\rho| = x^\beta/|\rho| \le x^{1/2}/|\rho|$ for $\beta \le 1$, and $\widehat{K}_H \ge 0$ is a scalar weight. □

### 9.4 Zero-free band

We can now exclude off-line zeros within the effective band provided the *finite* margin dominates the ledger, completion, and spectral envelope.

**Theorem 9.1** (Positivity barrier; zero-free band). *Assume ETI and the calibration of Lemma* **??**. *Suppose that*

$$M_H \;>\; C_\Gamma \;+\; x^{1/2}\,\mathsf{S}_H^{\mathrm{cert}}, \tag{60}$$

*for some $x \in [X, 2X]$, where $C_\Gamma$ is a uniform bound for $|\mathcal{G}_H(x)|$ (cf. (47)) and $\mathsf{S}_H^{\mathrm{cert}} \geq \mathsf{S}_H$ is a certified envelope. Then $\Xi^{\mathrm{fin}}$ has no zero with $\Re s \neq \frac{1}{2}$ and $|\Im(s-\frac{1}{2})| \leq 2c/H$.*

*Proof.* Assume to the contrary that $\rho_0 = \beta_0 + i\gamma_0$ is a zero in the band with $\beta_0 \neq \frac{1}{2}$. By Lemma 9.2 choose $x = x_{\rho_0}$ so that

$$\Re\Big(\frac{x^{\rho_0}}{\rho_0}\Big) \;\geq\; \kappa_{\mathrm{al}}\,\frac{x^{\beta_0}}{|\rho_0|} \;\geq\; \kappa_{\mathrm{al}}\,\frac{x^{1/2}}{|\rho_0|}.$$

Take the real part of the explicit formula (42) at this $x$:

$$T_H(x) \;=\; \mathrm{Main}_H(x) \;-\sum_\rho \widehat{K}_H\Big(\frac{\rho-\frac{1}{2}}{i}\Big)\Re\Big(\frac{x^\rho}{\rho}\Big) \;+\; \Re\,\mathcal{G}_H(x) \;+\; \Re\,\mathcal{E}_H^{\mathrm{fin}}(x).$$

Using Lemma **??**, $\mathrm{Main}_H(x) \leq c_0 + O(H^{-1})$, and by the ledger, $T_H(x) \geq c_0 - (C_1\varepsilon_H + C_2H^{-1} + C_3e^{-B}) = M_H$. Rearranging gives

$$\sum_\rho \widehat{K}_H\Big(\frac{\rho-\frac{1}{2}}{i}\Big)\Re\Big(\frac{x^\rho}{\rho}\Big) \;\leq\; (c_0 + O(H^{-1})) \;-\; M_H \;+\; C_\Gamma \;+\; O(\varepsilon_H + H^{-1} + e^{-B}).$$

Absorb the $O(\cdot)$ terms into $M_H$ (by its definition). Then

$$\sum_\rho \widehat{K}_H\Big(\frac{\rho-\frac{1}{2}}{i}\Big)\Re\Big(\frac{x^\rho}{\rho}\Big) \;\leq\; C_\Gamma \;-\; M_H.$$

Split off the term $\rho_0$ and bound the rest by Lemma 9.3:

$$\widehat{K}_H\Big(\frac{\rho_0-\frac{1}{2}}{i}\Big)\Re\Big(\frac{x^{\rho_0}}{\rho_0}\Big) \;+\; \sum_{\rho\neq\rho_0} \widehat{K}_H\Big(\frac{\rho-\frac{1}{2}}{i}\Big)\Big|\Re\Big(\frac{x^\rho}{\rho}\Big)\Big| \;\leq\; C_\Gamma \;-\; M_H.$$

By alignment and $\widehat{K}_H \geq 0$,

$$\kappa_{\mathrm{al}}\,\widehat{K}_H\Big(\frac{\rho_0-\frac{1}{2}}{i}\Big)\,\frac{x^{1/2}}{|\rho_0|} \;\leq\; C_\Gamma \;-\; M_H \;-\; x^{1/2}\,\mathsf{S}_H.$$

This contradicts (60) since $\kappa_{\mathrm{al}} \leq 1$ and $\mathsf{S}_H \leq \mathsf{S}_H^{\mathrm{cert}}$. ☐

*Remark* 9.1 (Role of PSD/coercivity). The theorem itself used only the nonnegativity and band-limitation of $\widehat{K}_H$ together with ETI. The PSD/coercivity from §8 is what makes $\mathsf{S}_H^{\mathrm{cert}}$ *computable* and stable: it controls the sampling of $|\Xi^{\mathrm{fin}}(\frac{1}{2} + i\xi)|$ and provides certified lower bounds on frame constants, ensuring the numerical argument-principle–type bound for $\mathsf{S}_H$ is reliable.

### 9.5 Dyadic rollout and certification

For a dyadic $[X, 2X]$ we certify: (i) FE to tolerance; (ii) PSD/coercivity (§8); (iii) margin $M_H$ from (57); (iv) the spectral envelope $\mathsf{S}_H^{\mathrm{cert}}$ via a fixed quadrature on the band; (v) a uniform bound $C_\Gamma$ for $\mathcal{G}_H$. If $M_H > C_\Gamma + x^{1/2}\mathsf{S}_H^{\mathrm{cert}}$ for some $x \in [X, 2X]$, the dyadic *passes* and Theorem 9.1 excludes off-line zeros of $\Xi^{\mathrm{fin}}$ in $|\Im(s - \frac{1}{2})| \le 2c/H$. Rolling over dyadics yields a ladder of certified bands; uniform ETI would globalize the conclusion.

## 10 Parameter Schedule and Safe Defaults

This section fixes concrete choices for the finite architecture parameters

$$(H, Q, c, B;\ M, h, \delta,\ J, \{w_j\}),$$

and records inequalities that guarantee all downstream analytic steps (ETI ledger transfer, band support, sampling/frame bounds, and positivity barrier) work uniformly on a dyadic window $[X, 2X]$. We tie everything to the *schedule exponents* $(A, \gamma)$ and band fraction $c$, as announced in (1).

### 10.1 Design constraints (summary)

The following constraints must be enforced:

(S1) **Band support.** The Fejér weight $\widehat{K}_H$ must be supported in $|t| \le 2c/H$ with $0 < c \le 1$; cf. (11).

(S2) **Bank resolution.** The bank size $M$ must resolve minor/major-arc structure: $M \asymp Q^2$; see §3.3.

(S3) **Ledger magnitude.** The ETI variance scale is $\varepsilon_H = (H/X)^{1/2} + (HQ^2)^{-1}$; cf. (25). The global ledger in the explicit formula is

$$\mathcal{L}_H \ \stackrel{\text{def}}{=}\ C_1\,\varepsilon_H \ +\ C_2\,H^{-1} \ +\ C_3\,e^{-B}. \tag{61}$$

(S4) **Sampling/frame.** The target grid step $h$ and jitter $\delta$ satisfy $h \le \pi H/4$ and $0 \le \delta < 1/4$ to secure the lower frame bound (7).

(S5) **Quadrature on the band.** The spectral quadrature $\{(\xi_j, w_j)\}_{j \le J}$ covers $\mathcal{B}_H = [-2c/H, 2c/H]$ with resolution $\asymp h^{-1}$; weights $w_j > 0$ sum to $2 \cdot (2c/H)$ up to a fixed constant.

(S6) **Margin dominance.** The positivity barrier needs

$$M_H \ =\ c_0 - \mathcal{L}_H \ >\ C_\Gamma + x^{1/2}\,\mathsf{S}_H^{\mathrm{cert}} \qquad \text{for some } x \in [X, 2X]. \tag{62}$$

### 10.2 Canonical schedule

Fix exponents and constants

$$A \ge 8, \qquad 0 < \gamma < \tfrac{1}{2}, \qquad 0 < c \le \tfrac{1}{2}, \qquad B \ge 12. \tag{63}$$

Let

$$H = (\log X)^A, \qquad Q = H^\gamma, \qquad N = 2X\,e^{B/H}. \tag{64}$$

Choose

$$M = \lfloor c_Q\,Q^2 \rfloor, \quad 1 \le c_Q \le 4; \qquad h = \frac{\pi H}{8}, \qquad 0 \le \delta \le \tfrac{1}{8}. \tag{65}$$

For spectral quadrature, take a uniform grid on $\mathcal{B}_H$ with spacing $\Delta\xi = \pi/H$ and equal weights $w_j = \Delta\xi$; then $J \asymp H^{-1}|\mathcal{B}_H|^{-1} \asymp H/c$.

### 10.3 Ledger asymptotics under the canonical schedule

With (64), the ETI scale reads

$$\varepsilon_H \;=\; \Big(\frac{H}{X}\Big)^{1/2} \;+\; H^{-1-2\gamma}\,. \tag{66}$$

Hence

$$\mathcal{L}_H \;=\; C_1\Big(\frac{H}{X}\Big)^{1/2} \;+\; C_1\,H^{-1-2\gamma} \;+\; C_2\,H^{-1} \;+\; C_3\,e^{-B}. \tag{67}$$

As $X \to \infty$ with fixed $(A, \gamma, B)$, the first term decays super-polynomially in $\log X$, while the remaining terms are $O(H^{-1}) = (\log X)^{-A}$. In particular,

$$\mathcal{L}_H \;\ll\; (\log X)^{-A} \;+\; e^{-B}. \tag{68}$$

### 10.4 Safe default numbers

For definiteness we adopt the following *safe defaults* (used in numerical certificates unless stated otherwise):

$$(A, \gamma, c, B) \;=\; (10,\, 0.30,\, \tfrac{1}{2},\, 12), \qquad c_Q = 2, \quad h = \frac{\pi H}{8}, \quad \delta = \frac{1}{8}. \tag{69}$$

Then

$$\mathcal{L}_H \;\le\; C_1\Big(\frac{(\log X)^{10}}{X}\Big)^{1/2} \;+\; C_1(\log X)^{-16} \;+\; C_2(\log X)^{-10} \;+\; C_3 e^{-12}. \tag{70}$$

For $X \ge 10^{12}$ the dominant pieces are the $(\log X)^{-10}$ tail and the fixed $e^{-12} \approx 6.1 \times 10^{-6}$; both are negligible for moderate $c_0$ (see below).

### 10.5 Checklist for barrier feasibility

Given ETI constants $(c_0, \varepsilon_H, C_{AO,SSU}, m_H)$ on $[X, 2X]$, the barrier (62) is feasible provided the following *deterministic* checks pass:

(C1) **Sampling/frame.** With (65) the frame inequality (7) holds with a constant $C_{\mathrm{frame}} \asymp h(1-4\delta)$.

(C2) **Band width.** The Fejér band is $|\xi| \le 2c/H$, so alignment error is $1 - \kappa_{\mathrm{al}} = O(H^{-2})$ (Lemma 9.2).

(C3) **Completion.** With $\mathbf{G}_H(s) = \exp(-(s - \frac{1}{2})^2/(2\alpha_H^2))$, the gamma term is bounded by a universal $C_\Gamma$ (see (47)).

(C4) **Ledger.** Compute $\mathcal{L}_H$ via (61); verify $\mathcal{L}_H \le \theta\, c_0$ with $\theta \in (0,1)$, e.g. $\theta = 1/4$.

(C5) **Envelope.** Bound $\mathsf{S}_H$ by a certified envelope $\mathsf{S}_H^{\mathrm{cert}}$ from the PSD kernel sampling (§8).

(C6) **Margin test.** Choose $x \in [X, 2X]$ maximizing the LHS phase alignment (Lemma 9.2); then check (60).

### 10.6 Sensitivity and tuning

- Increasing $A$ tightens all $(\log X)^{-A}$ tails and reduces the band width $O(1/H)$; it also increases quadrature/sample costs $\asymp H$.
- Decreasing $\gamma$ weakens the $1/(HQ^2)$ component; values $\gamma \in [0.2, 0.4]$ are a robust window.
- Smaller $c$ narrows the band, improving alignment and reducing spectral mass, at the cost of more dyadic steps to sweep a fixed height interval.
- Larger $B$ exponentially suppresses the finite Dirichlet tail and is essentially free computationally (we set $B \geq 12$ by default).

### 10.7 A quantitative target for certificates

For numerical rollout we adopt the quantitative target

$$\mathcal{L}_H \;\leq\; \frac{c_0}{4}, \qquad C_\Gamma \;\leq\; \frac{c_0}{8}, \qquad x^{1/2}\,\mathsf{S}_H^{\mathrm{cert}} \;\leq\; \frac{c_0}{8} \quad \text{for some } x \in [X, 2X]. \tag{71}$$

Under (71) one has $M_H \geq c_0/2$, and the margin inequality (62) holds automatically, so Theorem 9.1 gives a certified zero-free band for $\Xi^{\mathrm{fin}}$ on $|\Im(s - \frac{1}{2})| \leq 2c/H$.

### 10.8 Minimal schedule for theory-only guarantees

When numerical certification is not attempted, one may take a *coarser* but theory-safe schedule:

$$(A, \gamma, c, B) \;=\; (8,\, 0.25,\, \tfrac{1}{2},\, 12), \qquad h = \frac{\pi H}{10}, \quad \delta = \frac{1}{10}, \quad c_Q = 1. \tag{72}$$

All analytic lemmas and the explicit formula remain valid with slightly weaker constants; the ledger still satisfies $\mathcal{L}_H \ll (\log X)^{-8} + e^{-12}$.

**Finite-conditional acceptance checklist (per dyadic).** Given ETI constants and the schedule of §10, accept (pass) if:

(AC1) FE diagnostic: $\max_\xi |\Xi_{\mathrm{fin}}(\frac{1}{2} + i\xi) - \Xi_{\mathrm{fin}}(\frac{1}{2} - i\xi)| \leq 100\,\varepsilon_{\mathrm{mach}} \max_\xi |\Xi_{\mathrm{fin}}|$.

(AC2) PSD: Cholesky pivots of $\mathbf{K}$ satisfy $\geq -\tau_{\mathrm{PSD}}$; optional coercivity $\lambda_{\min}(\mathbf{K}) \geq \frac{1}{2}(m_H^2/2\pi)C_{\mathrm{frame}}$.

(AC3) Envelope: obtain $\mathsf{S}_H^{\mathrm{cert}}$ (TBA) or use the PSD-based proxy bound of §15.

(AC4) Barrier: $M_H > C_\Gamma + x^{1/2}\mathsf{S}_H^{\mathrm{cert}}$ for some $x \in [X, 2X]$.

Passing (AC1)–(AC4) yields the finite-conditional zero-free band (Theorem 1.1).

## Ledger constants (analytic decomposition)

We decompose

$$\mathcal{L}_H \;=\; C_1\,\varepsilon_H \;+\; C_2\,H^{-1} \;+\; C_3\,e^{-B}.$$

Here $C_1$ multiplies the ETI variance scale $\varepsilon_H$ and is arithmetic; $C_2, C_3$ are purely analytic.

| Constant | Source inequality | Conservative bound (fixed kernels) |
|---|---|---|
| $C_2$ | Transfer $T_H - \Psi_{H,Q}$ and band tails | $C_2 \le \|zK_H(z)\|_{L^1(\mathbb{R})} + \|\widehat{K_H} - \widehat{J_H}\|_{L^\infty} \cdot \|b\|_{\ell^2}/H^{1/2}$ |
| $C_3$ | Finite cutoff $N = 2Xe^{B/H}$ in Dirichlet tails | $C_3 \le \sup_{\sigma\ge 1+\delta}\ \|K_H\|_{L^1} \cdot \left(\sum_{n>N} \frac{\Lambda(n)}{n^\sigma}\right) \cdot e^{B/H}$ |

With the Fejér band kernel $K_H$ (unit mass, compact Fourier support) and Mellin-even Gaussian $J_H$, $C_2 = O(1)$ and $C_3 = O(1)$ depend only on the fixed kernel family and $\delta$; neither depends on ETI. Only $C_1\varepsilon_H$ is ETI-driven.

# 11 Certificates and Reproducibility (TBA)

**Finite-conditional status.** Throughout Draft 2, every "certificate" (§11) is prospective (TBA). All conclusions are *finite-conditional*: they are rigorously deduced from ETI$(X; H, Q)$, the explicit formula with nonnegative weights (§7), the PSD kernel and coercivity (§8), and the positivity barrier (§9). No single-shift hypotheses (e.g. $\Delta = 2$) are used; all arithmetic enters via ETI short-shift averages and bank balancing. The paper's diagnostic object and pass/fail verdicts (§13) are therefore *conditional on ETI + schedule*, independent of unproved global statements about $\zeta$.

*Status.* No machine-checked certificates are shipped with this draft. All certificate artifacts described below are **TBA** and will be generated only after a verified machine-checking toolchain is in place. Until then, the results in this paper are purely analytical; any references to certificates serve as a roadmap and do not constitute evidence.

## 11.1 Scope of intended certificates (design only)

When the toolchain is ready, each dyadic window $[X, 2X]$ will be accompanied by the following artifacts, each with a content hash and a manifest:

(C1) **FE** (Functional Equation) certificate: numerical verification that the completed finite object $\Xi^{\text{fin}}$ satisfies $\Xi^{\text{fin}}(1-s) = \Xi^{\text{fin}}(s)$ to tolerance $\tau_{\text{FE}}$ on a test grid that $H$-resolves the band.

(C2) **PSD** certificate: numerical proof that the discrete kernel matrix $\mathbf{K}$ in §8 is positive semidefinite to tolerance $\tau_{\text{PSD}}$; includes $\lambda_{\min}(\tilde{\mathbf{K}})$ and conditioning diagnostics.

(C3) **Coercivity/Frame** certificate: lower frame constant $C_{\text{frame}}$ for the chosen grid $(h, \delta)$ and a certified bound on $m_H$ (band nonvanishing), yielding a numerical coercivity constant $\kappa_H^{\text{num}}$ (cf. Theorem 8.1).

(C4) **Margin** certificate: explicit evaluation of

$$M_H = c_0 - C_1\varepsilon_H - C_2 H^{-1} - C_3 e^{-B}$$

with all components recorded and a pass/fail flag for $M_H > 0$.

(C5) **Spectral envelope** certificate: an upper bound $\mathsf{S}_H^{\text{cert}} \ge \mathsf{S}_H$ based on argument-principle sampling of $\Xi^{\text{fin}}(\frac{1}{2} + i\xi)$ on the band (see §8 and §9).

(C6) **Schedule** certificate: the concrete schedule used $(A, \gamma, c, B, M, h, \delta, J)$ and derivation of the ledger $\mathcal{L}_H$ in (61).

### 11.2 Determinism, rounding, and hashing (policy, TBA)

All future certificates will be produced by deterministic code with:

- fixed summation order (pairwise/Kahan) and fixed rounding mode (IEEE-754, round-to-nearest);
- integer or rational summation for combinatorial tallies where feasible;
- seeded PRNG for any randomized quadrature (seed recorded in manifest);
- SHA-256 (or stronger) content hashes for all inputs and outputs, including parameter JSON and grids;
- CPU model, OS, compiler/BLAS version and compile flags recorded for reproducibility.

These rules are *binding* once certificates are enabled; until then, they are design intent only.

### 11.3 Manifest & schema stubs (illustrative; TBA)

The following schemas are *illustrative only*; files with these shapes will be produced once machine checking is available. They are *not* present in this draft.


```
{
  "altzeta_version": "draft2",
  "dyadic": {"X": "...", "H": "...", "Q": "..."},
  "schedule": {"A": 10, "gamma": 0.30, "c": 0.5, "B": 12,
               "M": "...", "h": "...", "delta": "...", "J": "..."},
  "kernels": {"JH": "Mellin-even Gaussian", "KH": "Fejér band", "cpar": 0.5},
  "calibration": {"eta_HQ": "..."},
  "hashes": {"coeffs": "sha256:...", "grids": "sha256:...", "code": "sha256:..."},
  "certificates": {
    "FE": {"tau_FE": 1e-12, "grid": "...", "status": "TBA"},
    "PSD": {"tau_PSD": 1e-12, "lambda_min": null, "status": "TBA"},
    "FRAME": {"C_frame": null, "m_H": null, "kappa_num": null, "status": "TBA"},
    "MARGIN": {"c0": null, "epsH": null, "ledger": null, "status": "TBA"},
    "S_ENVELOPE": {"S_H_cert": null, "method": "arg-principle + PSD", "status": "TBA"}
  }
}
```


### 11.4 What the barrier would certify (once available)

Given certified values with the quantitative target (71), the margin inequality (62) holds and Theorem 9.1 certifies that $\Xi^{\mathrm{fin}}$ has no zero with $\Re s \neq \frac{1}{2}$ and $|\Im(s - \frac{1}{2})| \leq 2c/H$ on this dyadic. At present, this statement is *conditional* on the ETI constants and the design intent of the certificate pipeline; no machine-checked evidence is provided in this draft.

### 11.5 Disclosure

All references to "certificate" in other sections should be read as **prospective**. Where a proof or bound says "by certificate" we include, in this draft, the corresponding *analytic* justification; the certificate hook is a placeholder for later reproducible validation.

### 11.6 TBA checklist for enabling certificates

(T1) Implement deterministic kernels and grids; lock summation order and rounding.

(T2) Implement the PSD kernel builder and eigenvalue bound with interval arithmetic.

(T3) Implement the argument-principle sampler for $\Xi^{\text{fin}}$ on the band and cross-check by symmetry.

(T4) Implement manifest and hashing; produce JSON stubs for each dyadic.

(T5) Add unit tests and cross-platform checks; publish scripts and logs.

## 12 Robustness and Sensitivity

This section quantifies how the main objects and the positivity barrier respond to small perturbations of kernels, masks, schedule parameters, and ETI constants. The goal is to make explicit the *continuity margins* within which all conclusions remain valid.

### 12.1 Perturbations of kernels and masks

Let $K_H$ and $J_H$ be the Fejér and Gaussian-completion kernels of §3, and let $\delta K$ be a log-line kernel with $\widehat{\delta K}$ supported in the band $\operatorname{supp}(\widehat{K_H})$ and $\|\widehat{\delta K}\|_\infty \le \eta$. Likewise let $\tilde{v}$ be any cyclic shift of a $\delta = 2$ mask $v$ or a perturbation obeying $\|\tilde{v} - v\|_2 \le \eta$.

**Lemma 12.1** (Kernel/mask stability for $T_H$)**.** *With $T_H$ as in (39) and $b = b_{H,Q}$ fixed, replacing $K_H$ by $K_H + \delta K$ and $v$ by $\tilde{v}$ changes the smoothed prime functional by*

$$\sup_{x\in[X,2X]} \left|T_H^{\text{new}}(x) - T_H(x)\right| \;\ll\; \eta\,(\log N) \;+\; \eta\,\|b\|_{\ell^2[2,N]}\,H^{-1/2},$$

*and the masked dispersion functional (22) by $O(\eta)$ in $L^2(dx/x)$.*

*Sketch.* Plancherel on the log-line and $\|\widehat{\delta K}\|_\infty \le \eta$ control the convolution; the $\tilde{v}$ change preserves the $\delta = 2$ constraints so AO/SSU survives with a multiplicative $(1+O(\eta))$ factor (cf. Lemma 5.3). □

**Lemma 12.2** (PSD kernel stability)**.** *Let $\mathbf{K}$ be the band kernel matrix of (50). If $K_H$ is replaced by $K_H + \delta K$ with $\|\widehat{\delta K}\|_\infty \le \eta$ and $\Xi^{\text{fin}}$ is replaced by $\Xi^{\text{fin}} + \delta\Xi$ with $\sup_{|\xi|\le 2c/H} |\delta\Xi(\tfrac12 + i\xi)| \le \eta\,\|\Xi^{\text{fin}}\|_\infty$, then*

$$\|\delta\mathbf{K}\|_2 \;\ll\; \eta\,\|\mathbf{K}\|_2.$$

*In particular, by Weyl's inequality, $|\lambda_{\min}(\mathbf{K}^{\text{new}}) - \lambda_{\min}(\mathbf{K})| \le \|\delta\mathbf{K}\|_2$.*

### 12.2 Sensitivity of $\Xi^{\text{fin}}$ to coefficient errors

Let $b = b_{H,Q}$ and let $\tilde{b} = b + \delta b$ with $\|\delta b\|_{\ell^2[2,N]}$ small. Define $\tilde{B}_H^{\text{fin}}$ and $\tilde{\Xi^{\text{fin}}}$ accordingly.

**Lemma 12.3** (Coefficient-to-object Lipschitz)**.** *For $\sigma \ge 1 + \delta_0 > 1$,*

$$\left|B^{\text{fin}}(s) - \tilde{B}_H^{\text{fin}}(s)\right| \;\le\; \Big(\sum_{n\le N} \frac{|\delta b(n)|^2}{n^{2\sigma}}\Big)^{1/2} \Big(\sum_{n\le N} 1\Big)^{1/2} \;\ll\; \|\delta b\|_{\ell^2}\,N^{1-\sigma},$$

*and, with the same completion $\mathbf{G}_H$,*

$$\left|\log\frac{\Xi^{\text{fin}}(s)}{\tilde{\Xi^{\text{fin}}}(s)}\right| \;\le\; c(\delta_0)\,\|\delta b\|_{\ell^2}\,N^{1-\sigma}, \qquad \left|\frac{\Xi^{\text{fin}}}{\tilde{\Xi^{\text{fin}}}}(s) - 1\right| \;\ll\; \|\delta b\|_{\ell^2}\,N^{1-\sigma}.$$

*Sketch.* Cauchy–Schwarz and the identity $\log \Xi^{\mathrm{fin}}(s) - \log \Xi^{\widetilde{\mathrm{fin}}}(s) = \frac{1}{2}\int_{1/2}^{s}(B^{\mathrm{fin}} - \tilde{B}_H^{\mathrm{fin}})(u) - (\cdot)(1 - u)\,du$. ☐

## 12.3 Barrier stability under perturbations

Recall the margin $M_H$ in (57) and the barrier condition $M_H > C_\Gamma + x^{1/2}\,\mathsf{S}_H^{\mathrm{cert}}$ from Theorem 9.1.

**Proposition 12.1** ($\varepsilon$-robust barrier)**.** *Fix $\varepsilon > 0$. If the strict margin*

$$M_H \;\geq\; C_\Gamma \;+\; x^{1/2}\,\mathsf{S}_H^{\mathrm{cert}} \;+\; 2\varepsilon \tag{73}$$

*holds for some $x \in [X, 2X]$, then the conclusion of Theorem 9.1 is stable under any collection of perturbations obeying*

$$\Delta_{\mathrm{ledger}} \;+\; \Delta_\Gamma \;+\; x^{1/2}\,\Delta_{\mathrm{env}} \;\leq\; \varepsilon,$$

*where $\Delta_{\mathrm{ledger}}$ is the change in $C_{1H}+C_2H^{-1}+C_3e^{-B}$ due to schedule/kernels, $\Delta_\Gamma$ is the change in the completion bound, and $\Delta_{\mathrm{env}}$ is the increase of $\mathsf{S}_H^{\mathrm{cert}}$ induced by PSD/kernel/coefficient perturbations.*

*Proof.* All three quantities enter the inequality in Theorem 9.1 additively; a gap of $2\varepsilon$ survives any cumulative increase of size $\leq \varepsilon$. ☐

## 12.4 Parameter sensitivities

Write $H = (\log X)^A$, $Q = H^\gamma$, and recall $\mathcal{L}_H = C_1[(H/X)^{1/2} + 1/(HQ^2)] + C_2H^{-1} + C_3e^{-B}$. Treating $X$ as fixed on a dyadic:

**W.r.t.** $A$. Since $H = e^{A\log\log X}$,

$$\frac{\partial \mathcal{L}_H}{\partial A} \;=\; \log\log X \cdot \Big[\frac{C_1}{2}\Big(\frac{1}{HX}\Big)^{1/2} \;-\; C_1(1+2\gamma)\,H^{-2-2\gamma} \;-\; C_2\,H^{-2}\Big] \;+\; O(e^{-B}).$$

Thus increasing $A$ decreases $\mathcal{L}_H$ for large $X$ (all bracketed terms are $O(H^{-1})$).

**W.r.t.** $\gamma$. Since $Q = H^\gamma$ and $\partial(1/(HQ^2))/\partial\gamma = -2(\log H)/(HQ^2)$,

$$\frac{\partial \mathcal{L}_H}{\partial \gamma} \;=\; -2C_1\,\frac{\log H}{HQ^2} \;<0,$$

so increasing $\gamma$ decreases the ledger term $1/(HQ^2)$ linearly in $\log H$.

**W.r.t.** $c$. The band half-width is $2c/H$. Reducing $c$ shrinks the spectral support (improves the envelope) and improves phase alignment (Lemma 9.2); it has no direct appearance in $\mathcal{L}_H$.

**W.r.t.** $B$. $\partial\mathcal{L}_H/\partial B = -C_3e^{-B}$; taking $B \geq 12$ makes this term negligible.

### 12.5 Numerical sensitivities and fuses

(N1) **PSD fuse.** If $\lambda_{\min}(\tilde{\mathbf{K}}) < \tau_{\mathrm{PSD}}$ (cf. (56)) after kernel/coeff perturbations, enlarge $J$ or tighten $h$ until the bound recovers.

(N2) **Frame fuse.** If the lower frame bound degrades under jitter $\delta$, reduce $\delta$ or $h$; $C_{\mathrm{frame}} \asymp h(1-4\delta)$ gives a quantitative knob.

(N3) **Envelope fuse.** If $x^{1/2}\mathsf{S}_H^{\mathrm{cert}}$ rises near $M_H$, reduce $c$ (narrower band), increase $A$ (larger $H$), or raise $B$; all decrease the RHS or increase the LHS margin.

(N4) **Completion fuse.** A larger $\alpha_H^{-1} = H$ narrows the Gaussian completion and reduces $C_\Gamma$.

### 12.6 Takeaway

All structural steps (ETI transfer, explicit formula with sign/weights, PSD kernel, coercivity, barrier) are *continuous* in the inputs. A strict margin gap (73) makes the zero-free band resistant to small implementation errors, parameter drift, and benign kernel/mask changes. The knobs with the largest leverage are $(A, \gamma, c)$ and the quadrature/frame settings $(h, \delta, J)$.

## 13 Rollout Protocol for Dyadic Bands

This section specifies a concrete, repeatable *rollout* for testing zero-freeness of the completed finite object $\Xi^{\mathrm{fin}}$ on successive dyadic windows $[X, 2X]$. The protocol ties together the ETI inputs (§5.1), the explicit formula with sign/weights (§7), the band PSD/coercivity (§8), the finite positivity barrier (§9), and the parameter schedule (§10). Certificates are *prospective* (TBA; see §11); until machine checking is available, the "certificate" steps below are placeholders.

### 13.1 Objectives and scope

Given a dyadic $[X, 2X]$ and a schedule $(H, Q, c, B)$, the rollout produces a *pass/fail* verdict for the band

$$\mathcal{B}_H = \{ \tfrac{1}{2} + i\xi : \ |\xi| \le 2c/H \},$$

namely: "no off-line zeros of $\Xi^{\mathrm{fin}}$ in $\mathcal{B}_H$." If the band passes for all dyadics in a ladder $X_0, 2X_0, 4X_0, \ldots$, then the resulting stack of zero-free bands covers increasing symmetric height intervals around the critical line. (Global, unconditional RH would require a uniform ETI across all dyadics.)

### 13.2 Per-dyadic pipeline (deterministic checklist)

Fix $(A, \gamma, c, B)$ and let $H, Q, N$ be set by (64). For the window $[X, 2X]$ do:

**P1. Set the schedule.** Choose $H = (\log X)^A$, $Q = H^\gamma$, $N = 2Xe^{B/H}$ and the band fraction $c$ as in §10. Fix $M, h, \delta, J, \{w_j\}$ per (65).

**P2. Fix kernels and bank.** Use the Mellin-even Gaussian completion $\mathbf{G}_H(s)$ from (18) and the Fejér log-band kernel $K_H$ from (10)–(11). Choose a $\delta = 2$ mask $v$ (Lemma 3.4) and the short-shift bump $\omega_H$ (29).

**P3. Build finite coefficients.** Form $b_{H,Q}(n)$ from (16) and the corresponding Dirichlet polynomial $B^{\mathrm{fin}}$ (17). Set $\eta_{H,Q}$ by the level calibration in Lemma **??**.

**P4. Complete.** Define $\Xi^{\mathrm{fin}}$ by (19). (Optionally "FE certificate" TBA in §11.)

**P5. Prime-side functional.** Form $T_H(x)$ (39) and transfer the ETI pin from $\Psi_{H,Q}$ via Lemma 7.1.

**P6. Ledger and margin.** Compute the ledger $\mathcal{L}_H$ as in (61), then the margin $M_H = c_0 - \mathcal{L}_H$ (57). (Here $c_0, _H$ are ETI inputs.)

**P7. Band PSD kernel.** Build the discrete kernel $\mathbf{K}$ (50) on $\mathcal{B}_H$ from sampling nodes $\{\xi_j\}$ and atoms $\Phi_k = \widehat{K}_H(\cdot - t_k)$; verify PSD and estimate coercivity $\kappa_H$ (Theorem 8.1). ("PSD/FRAME certificate" TBA.)

**P8. Spectral envelope.** Bound the envelope $\mathsf{S}_H = \sum_\rho \widehat{K}_H((\rho - \frac{1}{2})/i)/|\rho|$ restricted to $\mathcal{B}_H$ by a certified $\mathsf{S}_H^{\mathrm{cert}}$ using the PSD-based sampling/argument-principle routine (design in §11).

**P9. Phase alignment.** Choose $x \in [X, 2X]$ (deterministically, e.g. $x = X\, e^{\theta \in [\log X, \log(2X)]} \cos(\gamma\theta)$) to exploit Lemma 9.2; track $\kappa_{\mathrm{al}} = 1 - O(H^{-2})$.

**P10. Barrier test.** Check the margin inequality

$$M_H \;>\; C_\Gamma \;+\; x^{1/2}\, \mathsf{S}_H^{\mathrm{cert}}\,,$$

with $C_\Gamma$ the completion bound from (47). If it holds, Theorem 9.1 yields a *pass* (no off-line zeros on $\mathcal{B}_H$); else *fail*.

### 13.3 Definition: the alt–zeta diagnostic object

A *diagnostic object at* $[X, 2X]$ consists of the tuple

$$\mathcal{D}_X = \Big(H, Q, c, B;\; K_H, J_H;\; \text{bank } (M, v),\; \text{bump } \omega_H;\; \eta_{H,Q};\; \text{grids } (t_k), (\xi_j, w_j)\Big)$$

together with the derived quantities

$$\Xi_{\mathrm{fin}}, \quad T_H(x), \quad \mathbf{K}, \quad M_H,\; C_\Gamma,\; \mathsf{S}_H^{\mathrm{cert}}.$$

It *passes* if

$$\boxed{M_H \;>\; C_\Gamma \;+\; x^{1/2}\, \mathsf{S}_H^{\mathrm{cert}} \quad \text{for some } x \in [X, 2X]}$$

and the non-certifying numerics satisfy the sanity checks in §15. A pass implies the zero-free conclusion of Theorem 1.1 on the band $\mathcal{B}_H$. The manifest format is outlined in Appendix **??** (TBA).

### 13.4 Roll over dyadics (ladder policy)

Start at $X = X_0$ (large enough that ledger tails (68) are tiny) and iterate $X \leftarrow 2X$:

- Maintain $(A, \gamma, c, B)$; $H, Q$ update automatically via (64).
- Keep $h, \delta$ fixed relative to $H$ (e.g. $h = \pi H/8$, $\delta = 1/8$) to preserve frame bounds.
- Recompute $M_H$ and $\mathsf{S}_H^{\mathrm{cert}}$ on each dyadic; declare pass/fail as above.

A *rollout report* is the list of dyadics, their pass/fail status, margins $M_H$, and envelope values $x^{1/2}\mathsf{S}_H^{\mathrm{cert}}$.

### 13.5 Tuning and fallback (if fail)

If a dyadic fails the barrier inequality:

(F1) *Narrow the band*: decrease $c$ to reduce $x^{1/2}\mathsf{S}_H^{\mathrm{cert}}$ and improve alignment (Lemma 9.2).

(F2) *Sharpen the ledger*: increase $A$ (larger $H$), or increase $B$ to kill $e^{-B}$.

(F3) *Strengthen sampling*: decrease $h$ or $\delta$ to raise $C_{\mathrm{frame}}$ and sharpen PSD coercivity.

(F4) *Recalibrate*: adjust $\eta_{H,Q}$ within the bounded range to better match the main level (Lemma **??**).

By Proposition 12.1, any strict gap $M_H \geq C_\Gamma + x^{1/2}\mathsf{S}_H^{\mathrm{cert}} + 2\varepsilon$ is robust to small cumulative increases of the right-hand side up to $\varepsilon$.

### 13.6 Complexity and scaling

Under the canonical schedule (§10):

$$H = (\log X)^A, \quad Q = H^\gamma, \quad M \asymp Q^2, \quad J \asymp H/c,$$

so forming $\mathbf{K}$ costs $O(JM)$ kernel evaluations and $O(M^3)$ in a dense eigen-solve (typically mitigated by structure or iterative PSD checks). The ledger and main-term steps cost $O(N)$ with $N \asymp Xe^{B/H}$.

### 13.7 Reporting and manifests (TBA)

Each dyadic produces a manifest (TBA, §11) recording: $(X, H, Q, c, B)$, ETI constants $(c_{0,H}\,, C_{AO,SSU}, m_H)$, calibration $\eta_{H,Q}$, ledger $\mathcal{L}_H$, margin $M_H$, sampling parameters $(M, h, \delta, J)$, PSD and frame diagnostics, $C_\Gamma$, envelope bound $\mathsf{S}_H^{\mathrm{cert}}$, chosen $x$, and the pass/fail verdict.

### 13.8 Interpretation

A sustained sequence of passes over dyadics constitutes a certified stack of zero-free *finite* bands for $\Xi^{\mathrm{fin}}$. With a uniform-in-$X$ ETI and the limiting analysis of §6–§7, this program is intended as an *alt-zeta* route toward the classical zero-free strip. In this draft, the result is *conditional* on ETI and on the (TBA) certificates; see §11.

## 14 Future Modules and Extensions

This section outlines self-contained modules that extend the alt-zeta programme and strengthen the Draft 2 pipeline into a certifiable stack. Each module is specified by its interface (inputs/outputs), invariants, dependencies, and planned deliverables. All *certificate* references are prospective (TBA; see §11).

### 14.1 ETI Completion Pack (AO/SSU & Calibration)

**Goal.** Deliver an end-to-end generator for ETI constants $(c_{0,H}\,, C_{AO,SSU}, m_H)$ on a dyadic.

- **Inputs:** $(X; H, Q, c, B)$, kernels $(J_H, K_H)$, bank $\mathcal{B}$, mask $v$, bump $\omega_H$.
- **Outputs:** ETI constants; diagnostic logs for AO/SSU dispersion and pin stability.
- **Invariants:** $\delta{=}2$ balancing (§3.3); ledger scale (25).

- **Dependencies:** §3, §5.1.
- **Deliverables:** Deterministic calculators for PIN, AO/SSU, and $m_H$ (band nonvanishing proxy).

### 14.2 Goldbach–Twin Interface (ETI Sources)

**Goal.** Provide *finite* arithmetic sources for the pair term in $b_{H,Q}$ (cf. (16)).

- **Inputs:** Triple interface primitives (Goldbach $\Rightarrow$ short-shift pairs), Twin-prime extension (ETI-level).
- **Outputs:** $\Pi_H^{\mathrm{avg}}(n)$ with (15); variance and drift controls compatible with AO/SSU.
- **Assumption (this draft):** Goldbach (finite-base checked) and Twin (pending) provide ETI-scale bounds only.
- **Dependencies:** §4, §5.1.

### 14.3 Spectral Envelope Estimator

**Goal.** Bound $\mathsf{S}_H$ in (59) using band sampling and PSD coercivity.

- **Inputs:** $\Xi^{\mathrm{fin}}(\frac{1}{2} + i\xi)$ sampled on $\mathcal{B}_H$; kernel atoms $\Phi_k$; frame constants.
- **Outputs:** Upper bound $\mathsf{S}_H^{\mathrm{cert}} \geq \mathsf{S}_H$ with error bars; quadrature manifest (TBA).
- **Invariants:** Uses Theorem 8.1; obeys sampling regime (53).
- **Dependencies:** §8, §9.

### 14.4 PSD Kernel Factory & Checker

**Goal.** Construct $\mathbf{K}$ (§8) and certify PSD/coercivity numerically.

- **Inputs:** Grids $(t_k)$, $(\xi_j, w_j)$; $\Xi^{\mathrm{fin}}$ evaluations; $\widehat{K}_H$.
- **Outputs:** $\lambda_{\min}(\mathbf{K})$, conditioning diagnostics; $\kappa_H^{\mathrm{num}}$.
- **Invariants:** Interval arithmetic or directed rounding for eigen-bounds (TBA).
- **Dependencies:** §8, §10.

### 14.5 Argument–Principle Sampler (Band)

**Goal.** Count/locate zeros of $\Xi^{\mathrm{fin}}$ in $\mathcal{B}_H$ to assist envelope bounds and sanity checks.

- **Inputs:** $\Xi^{\mathrm{fin}}$ on rectangles enclosing $\mathcal{B}_H$; step controls tied to $1/H$.
- **Outputs:** Zero counts in band; stability flags; optional "no-zero drift" report.
- **Dependencies:** §4, §8.

### 14.6 Alternative Completions & Kernels (Robustness Lab)

**Goal.** Stress-test barrier under controlled changes to $(J_H, K_H)$.

- **Inputs:** Families of even, nonvanishing Mellin kernels; bandlimited $\widehat{K}$ variants.
- **Outputs:** Sensitivity plots for $M_H$, $\mathsf{S}_H^{\mathrm{cert}}$, and PSD constants.
- **Dependencies:** §12, §10.

### 14.7 Dirichlet Twists & GL(1) Family

**Goal.** Extend $B^{\mathrm{fin}}$ to twisted coefficients for primitive characters $\chi$.

- **Inputs:** $b_{H,Q}^{(\chi)}(n)$ via $\Lambda(n)\chi(n)$ and pair-weight analogues.
- **Outputs:** $\Xi^{\mathrm{fin}(\chi)}$ with finite FE and band explicit formula mirroring §7.
- **Dependencies:** §4, §6; schedule adapts with conductor.

### 14.8 GL(2) Testbed (Hecke/Eigenforms)

**Goal.** Prototype an alt-zeta analogue for $L(s, f)$ (holomorphic/newforms) with finite completion and band kernel.

- **Inputs:** Hecke eigenvalues $a_f(n)$; completed Gaussian factor adapted to gamma parameters of $f$.
- **Outputs:** $\Xi^{\mathrm{fin}(f)}$; PSD kernel on the band; barrier feasibility study.
- **Dependencies:** Generalization of §3–§8.

**Lemma 14.1** (Calibration monotonicity and optimality)**.** *Fix a dyadic $[X, 2X]$ and let $\Psi_{H,Q}$ be the $J_H$–smoothed ledger functional from §5, viewed as an element of $L^2([X, 2X], dx/x)$. Let $\mathcal{C}$ be the convex set of admissible calibrations (e.g. all $\eta_{H,Q}$ with prescribed linear constraints from §5) and write $\Psi_{H,Q}(\eta)$ for the calibrated version. Then the map*

$$\eta \longmapsto \left\|\Psi_{H,Q}(\eta) - c_0\right\|_{L^2(dx/x)}$$

*is strictly convex on $\mathcal{C}$, hence has at most one minimizer (the $L^2$–projection of $c_0$ onto the affine subspace $\Psi_{H,Q}(\mathcal{C})$). Moreover, if $\eta_1, \eta_2 \in \mathcal{C}$ and $\lambda \in [0, 1]$, then*

$$\left\|\Psi_{H,Q}(\lambda\eta_1 + (1-\lambda)\eta_2) - c_0\right\|_{L^2} \;\le\; \lambda\left\|\Psi_{H,Q}(\eta_1) - c_0\right\|_{L^2} + (1-\lambda)\left\|\Psi_{H,Q}(\eta_2) - c_0\right\|_{L^2}.$$

*In particular, any descent method on $\mathcal{C}$ decreases the ledger margin $M_H$ monotonically until it reaches the unique minimizer.*

*Proof.* By construction the calibration $\eta \mapsto \Psi_{H,Q}(\eta)$ is affine (linear in $\eta$ plus a fixed offset coming from the bank/mask split). The objective is the norm of an affine map in a Hilbert space; its square is a strictly convex quadratic unless the linear part is zero, which is excluded by the nontriviality of the calibration directions. Strict convexity implies uniqueness of the minimizer, which is the orthogonal projection of $c_0$ onto the image affine subspace. Jensen's inequality yields the displayed monotonicity. □

### 14.9 Formalization Hook (Lean/Isabelle)

**Goal.** Mechanize the *finite* theorems (FE, EF, PSD coercivity, barrier direction).

- **Inputs:** Typed algebra of kernels, Mellin/Fourier plumbing, finite Dirichlet objects.
- **Outputs:** Verified proofs of Theorems 4.1, 7.1, 8.1, 9.1 (finite scope).
- **Dependencies:** All preceding analytic sections; certificate placeholders remain TBA.

### 14.10 Numerical Reproducibility Pipeline

**Goal.** Enforce determinism and produce manifests (TBA; §11).

- **Inputs:** Parameter JSON, code hash, grids, rounding mode.
- **Outputs:** Manifests with SHA-256 hashes; logs for FE/PSD/envelope; pass/fail per dyadic.
- **Dependencies:** §11, §13.

### 14.11 Interfaces and Contracts (Summary)

Each module $M$ exposes:

| | |
|---|---|
| *Inputs:* | $(X, H, Q, c, B)$, kernels, bank/mask, and ETI hooks. |
| *Outputs:* | Numeric bounds (ledger, $m_H$, $\mathsf{S}_H^{\mathrm{cert}}$), PSD data, or finite objects $\Xi^{\mathrm{fin}(\cdot)}$. |
| *Invariants:* | Band support $\lvert\xi\rvert \le 2c/H$, $\delta = 2$ balancing, schedule safety (63). |
| *Deps:* | Explicit references to §3–§13. |

### 14.12 Risk Register (selected)

- **ETI under-specification.** Mitigation: restrict to ledger-scale claims; no single-shift usage (§5.1).
- **Envelope looseness.** Mitigation: reduce $c$, increase $A$, or refine PSD sampling (§12).
- **PSD conditioning.** Mitigation: adjust $(h, \delta)$ and $J$; use interval eigen-bounds (§8).
- **Calibration drift.** Mitigation: one-shot convex minimization for $\eta_{H,Q}$; freeze per dyadic (§**??**).

## 15 Numerical Methods and Implementation Plan

This section specifies concrete numerical procedures for all finite objects and tests used in the rollout (§13). We separate model choices (precision, rounding, determinism), core evaluations ($\Xi^{\mathrm{fin}}$ on the band and $T_H$), kernel assembly (PSD $\mathbf{K}$), and diagnostics (envelope, FE residual, ledgers). Certificates remain TBA (§11); where "certificate" appears below, read it as a reproducible computation hook.

### 15.1 Numerical model (deterministic by construction)

- **Precision.** Default IEEE-754 `binary64` (53-bit mantissa). Fallback to `binary128` or MPFR only for certificate-grade eigenvalue bounds and argument-principle loops.
- **Rounding & FMA.** Round-to-nearest ties-to-even; enable fused multiply–add (FMA) where available.
- **Summation.** Pairwise/Kahan for all long sums; stable $\ell^2$-norm via two-pass dot products.
- **Determinism.** Fixed iteration orders; no SIMD reordering affecting associativity; disable "fast-math". All grids and parameters are hashed into the manifest (TBA).

### 15.2 Evaluating the finite Dirichlet object on the band

We need $B^{\text{fin}}(s)$ and $\Xi^{\text{fin}}(s)$ for $s = \frac{1}{2} + i\xi$, $|\xi| \le 2c/H$.

(D1) **Coefficient cache.** Precompute $b_{H,Q}(n)$ from (16) and store $\{\log n,\, b_{H,Q}(n)\}_{2\le n\le N}$.

(D2) **Vector evaluation.** For a vector $\xi \in \mathbb{R}^J$, compute $n^{-s} = \exp(-s\log n)$ via $\exp(-\frac{1}{2}\log n)\,[\cos(\xi\log n) - i\sin(\xi\log n)]$, vectorized over $n$ and $\xi$.

(D3) **Stability.** Partition $[2,N]$ into dyadic blocks to keep $\xi\log n$ in moderate ranges; use remapping $\xi\log n \mapsto \xi\log n \mod 2\pi$ to reduce trigonometric loss.

(D4) **Completion.** Multiply by $\mathbf{G}_H(s) = \exp(-(s-\frac{1}{2})^2/(2\alpha_H^2))$ from (18); form $\Xi^{\text{fin}}$ via (19): a line integral of $(B^{\text{fin}}(u) - B^{\text{fin}}(1-u))/2$ from $u = 1/2$ to $s$, implemented with a Gauss–Legendre rule in the $u$-variable (odd integrand about $1/2$ gives cancellation).

(D5) **FE residual (diagnostic).** Compute $\text{FE\_res}(\xi) = \Xi^{\text{fin}}(\frac{1}{2}+i\xi) - \Xi^{\text{fin}}(\frac{1}{2}-i\xi)$; report $\max_\xi |\text{FE\_res}|$ as the non-certifying FE diagnostic.

### 15.3 Computing the smoothed prime functional $T_H(x)$

Recall $T_H(x) = \sum_{n\le N} b_{H,Q}(n)\,K_H(\log n - \log x)$ with $K_H$ from (10).

(P1) **Direct band-limited sum.** Since $K_H$ decays like $z^{-2}$ outside the compact Fourier band and we only query $x \in [X, 2X]$, restrict to $n$ with $|\log n - \log x| \le 10/H$.

(P2) **Grided convolution (optional).** Build a log-grid $z_m = \log X + m\Delta$ with $\Delta = \pi/H$; precompute $B_m = \sum_n b_{H,Q}(n)\,\mathbf{1}_{\{\log n\in[z_m-\Delta/2, z_m+\Delta/2)\}}$ and convolve $B$ with $K_H$ by FFT on the torus of length $\asymp \log 2$.

(P3) **Transfer check.** Compare $T_H(x)$ with $\Psi_{H,Q}(x)$ (Lemma 7.1); record $\|T_H - \Psi_{H,Q}\|_{L^2(dx/x;[X,2X])}$ to confirm $O(H^{-1})$ behavior.

**Theorem 15.1** (PSD coercivity on the band)**.** *Let $B_H = [-2c/H,\, 2c/H]$ and suppose $m_H := \inf_{\xi\in B_H}|\Xi_{\text{fin}}(\frac{1}{2}+i\xi)| > 0$. Let $\Phi_k(\xi) = K_{c,H}(\xi - t_k)$ with $K_{c,H} \ge 0$ even, $\operatorname{supp}\Phi_k \subset B_H$, and sampling $\{t_k\}$ obeying the schedule of §8 (step $h \le \pi H/4$, jitter $\delta < 1/4$). Then there exists a constant $C_{\text{frame}} > 0$ (depending only on the Fejér family and schedule) such that*

$$\sum_{k,\ell} \overline{a_k}\, a_\ell\, \frac{1}{2\pi}\int_{B_H} \Xi_{\text{fin}}\Big(\tfrac{1}{2}+i\xi\Big)\,\overline{\Xi_{\text{fin}}\Big(\tfrac{1}{2}+i\xi\Big)}\,\Phi_k(\xi)\,\Phi_\ell(\xi)\,d\xi \;\ge\; \frac{m_H^2}{2\pi}\,C_{\text{frame}}\sum_k |a_k|^2.$$

*Equivalently, for the discretized matrix $K = B^*WB$ assembled as in §15.4, $\lambda_{\min}(K) \ge (m_H^2/2\pi)\,C_{\text{frame}}$.*

*Proof.* Let $F(\xi) := \Xi_{\text{fin}}(\frac{1}{2} + i\xi)$ and note $|F(\xi)| \geq m_H$ on $B_H$. For any finitely supported $(a_k)$,

$$\sum_{k,\ell} \overline{a_k} a_\ell \frac{1}{2\pi} \int_{B_H} |F(\xi)|^2 \, \Phi_k(\xi)\Phi_\ell(\xi) \, d\xi = \frac{1}{2\pi} \int_{B_H} |F(\xi)|^2 \left| \sum_k a_k \Phi_k(\xi) \right|^2 d\xi.$$

Lower bound $|F|^2 \geq m_H^2$ and invoke the discrete Fejér frame inequality on $B_H$ (the schedule yields a stable frame for the Fejér atoms):

$$\int_{B_H} \left| \sum_k a_k \Phi_k(\xi) \right|^2 d\xi \;\geq\; C_{\text{frame}} \sum_k |a_k|^2.$$

Combining the two inequalities gives the claim in the continuous model. The discretized claim follows by standard quadrature stability (weights $w_j \simeq \Delta\xi = \pi/H$ on the same band and the compact support of $K_{c,H}$), which is exactly the (K1)–(K4) assembly used to build $K$. □

### 15.4 PSD kernel assembly and checks

We assemble $\mathbf{K} = \mathbf{B}^*\mathbf{W}\mathbf{B}$ with entries $\mathbf{B}_{jk} = \frac{1}{\sqrt{2\pi}} \Xi^{\text{fin}}(\frac{1}{2} + i\xi_j)\, \Phi_k(\xi_j)$ where $\Phi_k(\xi) = \widehat{K}_H(\xi - t_k)$ and $\widehat{K}_H$ is from (11).

(K1) **Sampling.** Choose $\{\xi_j\} \subset [-2c/H, 2c/H]$ with spacing $\Delta\xi = \pi/H$; set $w_j = \Delta\xi$. Choose $t_k$ with step $h \leq \pi H/4$ and jitter $\delta < 1/4$ (§8).

(K2) **Matrix build.** Precompute $X_j = \Xi^{\text{fin}}(\frac{1}{2} + i\xi_j)$. For each $k$, fill column $k$ by $B_{jk} = (2\pi)^{-1/2} X_j \, \widehat{K}_H(\xi_j - t_k)$. Exploit the compact support of $\widehat{K}_H$ to sparsify $\mathbf{B}$.

(K3) **PSD check.** Form $\mathbf{K} = \mathbf{B}^*\mathbf{W}\mathbf{B}$. Compute a Cholesky factorization with symmetric pivoting; accept PSD if all pivots $\geq -\tau_{\text{PSD}}$ (tolerance tied to machine epsilon).

(K4) **Coercivity proxy.** Estimate $\lambda_{\min}(\mathbf{K})$ by Lanczos (warm-start with multiple random vectors); compare to $\kappa_H^{\text{num}} = (m_H^2/2\pi) C_{\text{frame}}$ (Theorem 8.1).

(K5) **Interval fallback (TBA).** For certificates, bound $\lambda_{\min}$ using interval LDL$^\top$ or Gershgorin with compensated sums; record bounds and environment in the manifest.

### 15.5 Spectral envelope bound $\mathsf{S}_H^{\text{cert}}$ (operational recipe)

We bound $\mathsf{S}_H = \sum_\rho \widehat{K}_H((\rho - \frac{1}{2})/i)/|\rho|$ on the band.

(E1) **Zero census (TBA).** Use an argument–principle loop on the rectangle $\{\, \frac{1}{2} + i\xi : |\xi| \leq 2c/H \,\}$ extended by a small $\sigma$-buffer to count band zeros.

(E2) **Lower moduli.** Lower-bound $|\rho|$ within the band by $|\frac{1}{2} + i\xi|$ evaluated at the grid points closest to the counted zeros (worst-case allocation).

(E3) **Weight cap.** Since $0 \leq \widehat{K}_H \leq 1$, an admissible upper bound is $\mathsf{S}_H^{\text{cert}} \leftarrow \sum_{\text{counted zeros}} 1/|\rho|_{\min}$. If no census is available, use an a priori cap based on the band length and a maximal density model, then shrink via PSD coercivity:

$$\mathsf{S}_H^{\text{cert}} \;\leq\; \frac{1}{m_H^2} \int_{\mathcal{B}_H} |X(\xi)|^2 \, d\xi \;\cdot\; \sup_{t \in \mathbb{R}} \int \widehat{K}_H(\xi - t) \, d\xi,$$

where $X(\xi) = \Xi^{\text{fin}}(\frac{1}{2} + i\xi)$ (Cauchy–Schwarz plus $\widehat{K}_H \leq 1$). The integral is computed from $X_j$ and $w_j$.

### 15.6 Completion term bound $C_\Gamma$

Compute $\mathcal{G}_H(x)$ via (47) at quadrature nodes on $\Re s = 1+\varepsilon$; report $C_\Gamma = \sup_{x\in[X,2X]} |\mathcal{G}_H(x)|$. With the Gaussian completion (18), $(\log \mathbf{G}_H)'$ is linear, so a coarse $t$-grid suffices; bound tails by the compact $t$-support of $\widehat{K}_H$.

[Completion bound] Let $G_H$ be the completion factor from §4 and define

$$G_H(x) \;:=\; \frac{1}{2\pi i}\int_{\Re s=1+\varepsilon} (\log G_H)'(s)\,\frac{x^s}{s}\,K_{c,H}\left(\frac{s-\frac{1}{2}}{i}\right)\,ds,$$

with $K_{c,H}$ even, $0 \le K_{c,H} \le 1$, and ${}_tK_{c,H}(\frac{s-\frac{1}{2}}{i}) \subset \{|t| \le 2c/H\}$. Then for each fixed $\varepsilon \in (0,1)$ there exists a constant $C(\varepsilon)$ (depending only on the fixed Gaussian completion and $c$) such that

$$\sup_{x>0} |G_H(x)| \;\le\; C(\varepsilon).$$

If in addition $(\log G_H)'$ is affine in $t$ along vertical lines (as for the Mellin–Gaussian completion), then the stronger estimate $\sup_{x>0} |G_H(x)| \;\ll\; 1 + \frac{c^2}{H^2}$ holds uniformly in $H$.

*Proof.* Because $K_{c,H}$ is compactly supported in $t$, the vertical integration is over a finite interval of length $O(1/H)$. Shift the contour to $\Re s = -\varepsilon$; the only potential residue is at $s = 0$ (from the factor $1/s$). Since $K_{c,H}((s-\frac{1}{2})/i)$ is even and vanishes to first order at $s = 0$ (Fejér shape), and $(\log G_H)'$ is odd about $1/2$, the residue at $s = 0$ is a bounded constant independent of $x$. The two vertical integrals are absolutely convergent by the compact $t$–support and smoothness of $(\log G_H)'$, giving a uniform-in-$x$ bound. When $(\log G_H)'(1+\varepsilon+it)$ is affine in $t$, its size on the truncated interval is $O(1+|t|)$, so integrating against a kernel supported on $|t| \le 2c/H$ yields the quantitative $O(1 + c^2/H^2)$ bound. $\square$

### 15.7 Error budget and acceptance criteria

All barrier inputs appear additively in

$$M_H \overset{(57)}{=} c_0 - \Big(C_{1H} + C_2H^{-1} + C_3e^{-B}\Big) \quad \text{vs} \quad C_\Gamma + x^{1/2}\,\mathsf{S}_H^{\text{cert}}.$$

We recommend the following (non-certifying) acceptance thresholds for a dyadic pass:

| | | |
|---|---|---|
| FE residual | : | $\max_\xi |\Xi^{\text{fin}}(\frac{1}{2}+i\xi) - \Xi^{\text{fin}}(\frac{1}{2}-i\xi)| \le 100\,\varepsilon_{\text{mach}}\,\max_\xi |\Xi^{\text{fin}}|$ |
| PSD | : | all Cholesky pivots $\ge -\tau_{\text{PSD}}$ with $\tau_{\text{PSD}} = 100\,M\,\varepsilon_{\text{mach}}$ |
| Coercivity | : | $\lambda_{\min}(\mathbf{K}) \ge \frac{1}{2}\,(m_H^2/2\pi)\,C_{\text{frame}}$ |
| Envelope | : | $\mathsf{S}_H^{\text{cert}}$ from (E1)–(E3) |
| Barrier | : | $M_H > C_\Gamma + x^{1/2}\,\mathsf{S}_H^{\text{cert}}$. |

### 15.8 Complexity summary (canonical schedule)

With $H = (\log X)^A$, $Q = H^\gamma$, $M \asymp Q^2$, $J \asymp H/c$ (cf. §10):

$$\text{Build } b_{H,Q}:\ O(N), \qquad \Xi^{\text{fin}} \text{ on band}:\ O(J\,N), \qquad \mathbf{K} \text{ build}:\ O(JM), \qquad \text{eig.}:\ O(M^3)\ \text{(dense)}.$$

Sparsity from $\widehat{K}_H$ can reduce $JM$ substantially; PSD checks by Lanczos give $O(M^2 \cdot \text{iters})$.

### 15.9 Sanity checks

- **Kernel duality.** Numerically verify $\int K_H = 1$ and $\widehat{K}_H = \widehat{K}_H$ on the band to machine tolerance.
- **Transfer.** Check $\|T_H - \Psi_{H,Q}\|_{L^2(dx/x)} = O(H^{-1})$ (Lemma 7.1).
- **Oddness.** Verify $(\log \mathbf{G}_H)'(1/2 + i\xi) = -(\log \mathbf{G}_H)'(1/2 - i\xi)$ numerically.
- **Grid stability.** Perturb $h, \delta$ within the robustness envelope (§12); monitor $\lambda_{\min}(\mathbf{K})$ and the barrier inequality.

### 15.10 Outputs recorded (TBA manifests)

Per dyadic: hashes of $\{b_{H,Q}\}$, grid specs, FE residual, PSD pivots/eigs, $C_\Gamma$, $\mathsf{S}_H^{\text{cert}}$, $M_H$, barrier verdict, environment (CPU/BLAS/compiler). See §11.

### 15.11 Envelope bound without coercivity (numeric fallback)

Define the band rectangle

$$\mathcal{R}_H(\tau) \;=\; \{\, \sigma + i\xi : \tfrac{1}{2} - \sigma_0 \le \sigma \le \tfrac{1}{2} + \sigma_0,\; |\xi - \tau| \le 2c/H \,\},$$

with fixed $\sigma_0 \in (0, 1/4]$. By the argument principle,

$$N_H(\tau) \;=\; \frac{1}{2\pi i} \oint_{\partial\mathcal{R}_H(\tau)} \frac{\Xi_{\text{fin}}'(s)}{\Xi_{\text{fin}}(s)}\, ds$$

counts zeros (with multiplicity) in the translated band. This integral is *computable* from sampled values of $\Xi_{\text{fin}}$ and $\Xi_{\text{fin}}'$ on $\partial\mathcal{R}_H(\tau)$ and does not require $m_H$. Given $N_H(\tau)$, a crude but certified envelope bound is

$$\mathsf{S}_{H,\tau}^{\text{cert}} \;\le\; \frac{N_H(\tau)}{\min_{\rho \in \mathcal{R}_H(\tau)} |\rho|} \;\le\; \frac{N_H(\tau)}{\frac{1}{2}} \;=\; 2\, N_H(\tau), \tag{74}$$

since $|\rho| \ge \frac{1}{2}$ throughout the rectangle. A sharper purely numeric bound is obtained by locating each zero within the band to a small disk and summing $1/|\rho|$ with interval arithmetic. This path yields a *non-coercive*, ETI-free envelope estimate suitable for Theorem 9.1 (cf. Remark 8.2).

## 16 Discussion and Outlook

### 16.1 What this draft establishes (finite scope)

The pipeline developed here is finite, band-limited, and modular:

(D1) A completed finite object $\Xi^{\text{fin}}$ with an *exact* functional equation, engineered by a Mellin-even, nonvanishing completion (§4).

(D2) A smoothed explicit formula (§7) in which zeros enter with a *global minus sign* and *nonnegative*, band-limited weights $\widehat{K}_H$.

(D3) A de Branges–type *PSD kernel* on the effective band and a coercivity mechanism (§8) that turn sign/weights into quadratic-form nonnegativity.

(D4) A *finite positivity barrier* that excludes off-line zeros of $\Xi^{\text{fin}}$ on the band when the ETI pin/ledger dominate the completion and spectral envelope (§9).

(D5) A *rollout protocol* and *parameter schedule* that specify how to verify the barrier on successive dyadics (§13, §10).

All statements are conditional on the Extended Triple Interface (ETI; §5.1), which supplies only short-shift and band-limited inputs; no single-shift (e.g. twin) assertions are used directly. Certificates are prospective (TBA; §11).

## 16.2 Why alt-zeta?

Classically, one pays for analytic continuation and gamma factors; on the explicit-formula side, zero contributions arrive with mixed signs unless the test function is carefully engineered. Here we *choose* the completion to enforce evenness, we *choose* $K_H$ so that its Fourier multiplier $\widehat{K_H}$ is nonnegative and compactly supported, and we *choose* the finite coefficients $b_{H,Q}$ to interface with pair information through ETI. The result is a configuration expressly tuned for a PSD argument on the band.

## 16.3 Band semantics and height scanning

Throughout, the "band" refers to the frequency window $|\xi| \le 2c/H$ for $s = \frac{1}{2} + i\xi$; the Fejér multiplier $\widehat{K_H}$ localizes contributions in this variable. To probe zeros near an *arbitrary* height $T$, one introduces a *carrier shift*

$$K_{H,\tau}(z) \;:=\; e^{i\tau z}\, K_H(z) \qquad \Longleftrightarrow \qquad \widehat{K_{H,\tau}}(t) \;=\; \widehat{K_H}(t-\tau), \tag{75}$$

with $\tau \approx T$. The explicit formula (Theorem 7.1) then weighs zeros by $\widehat{K_H}(\gamma-\tau)$ (where $\rho = \beta + i\gamma$), i.e. in a band centered at $\gamma = \tau$. The PSD kernel (§8) and barrier (§9) adapt verbatim with $\widehat{K_H}(\cdot - \tau)$. This is the standard shift-invariance of Paley–Wiener spaces and provides the missing "height scan" needed for a global program.

## 16.4 Roadmap to an unconditional RH (what would suffice)

The finite barrier excludes off-line zeros of $\Xi^{\text{fin}}$ in *any* chosen band around height $\tau$ provided the margin dominates the envelope. To transport this to classical $\zeta$, two ingredients suffice:

(R1) **Uniform ETI.** A version of Assumption 5.2 with constants $(c_{0,H}\,, C_{AO,SSU}\,, m_H)$ bounded uniformly in the dyadic $X$ and *uniformly in the carrier* $\tau$ used in (75). In particular, $c_0$ bounded away from 0 and $_H \to 0$ along the canonical schedule.

(R2) **Vanishing envelope.** For each fixed $\tau$, an upper bound $\mathsf{S}^{\text{cert}}_{H,\tau} \to 0$ as $H \to \infty$, uniformly for $\tau$ in compact height ranges; equivalently, control of the band mass of zeros for $\Xi^{\text{fin}}$ tuned by (75).

Under (R1)–(R2), the barrier (§9) forbids any off-line zero in any finite height window; letting the window drift and $H \to \infty$ excludes off-line zeros globally for $\Xi^{\text{fin}}$. Finally, the finite-to-classical link (§6, Lemma 6.1) identifies the zero loci in the limit, yielding RH. In this draft, (R1)–(R2) are programmatic targets, not established facts.

### 16.5 Limitations and failure modes

- **Dependence on ETI.** The ledger (24) and AO/SSU (26) are *inputs*. If they fail at the canonical scale (25), the margin may be negative.
- **Envelope looseness.** Crude bounds on $\mathsf{S}_H$ can swamp the margin. The carrier shift and PSD coercivity (§8) are designed to keep envelopes small and computable; absent that, the barrier is inconclusive.
- **Finite vs. classical.** $\Xi^{\text{fin}}$ is *not* $\Xi$; the link in Lemma 6.1 holds on $\Re s > 1$ and in smoothed senses tied to $(H, B)$; the limiting step must be justified carefully.
- **Numerics.** Without certificates, numerical checks are heuristic; see §11 for the TBA plan and §15 for deterministic policies.

### 16.6 Comparison with classical strategies

Levinson/Conrey–type mollifiers optimize mean values to count critical zeros; de Branges spaces control entire functions with built-in positivity. Our route borrows the *positivity* of de Branges but achieves it by tailoring the completion and the band kernel to a finite, ETI-driven Dirichlet object. The upside is modularity and short-shift input; the cost is the ETI hypothesis and the need to manage envelopes and carriers.

### 16.7 Where to invest effort next

(N1) **ETI completion.** Finish the Extended Triple Interface (ETI) from Goldbach and twins at the *finite* level (§14.2, §14.1). In this draft we use only short-shift, bank-balanced consequences.

(N2) **Carrier rollout.** Implement the carrier-shifted Fejér kernel (75) and add $\tau$ to the schedule; prove that all lemmas are stable under $\tau$ (they are, up to re-centering).

(N3) **Envelope estimator.** Finalize the PSD-assisted argument–principle sampler (§14.3, §14.5) to make $\mathsf{S}_{H,\tau}^{\text{cert}}$ small and certified.

(N4) **Formalization.** Mechanize the finite theorems (FE, EF, PSD coercivity, barrier) in Lean/Isabelle (§14.9) to eliminate book-keeping errors.

### 16.8 Robustness recap

The barrier is continuous in all inputs (§12). A strict gap $M_H \geq C_\Gamma + x^{1/2}\mathsf{S}_H^{\text{cert}} + 2\varepsilon$ survives small cumulative degradations up to $\varepsilon$ (Proposition 12.1). The most effective knobs are: increase $A$ (narrow band, smaller tails), decrease $c$ (less spectral mass), increase $\gamma$ (shrink $1/(HQ^2)$), refine $(h, \delta)$ (frame constant up), and enlarge $B$ (kill $e^{-B}$).

### 16.9 Closing remark

The alt-zeta programme reorganizes the battlefield: it trades classical globality for a finite, band-limited, positivity-driven setting where short-shift arithmetic enters only through ETI. If ETI can be completed (even modestly) and the spectral envelope tightened by PSD-driven sampling, the barrier becomes a lever strong enough—at least in principle—to exclude off-line zeros globally by carrier scanning. The remaining work is arithmetic (ETI), spectral (envelopes), and certification (deterministic numerics), not conceptual surgery on the analytic skeleton developed here.

# A Appendix A: Selected Proofs

### Functional equation (full detail)

Let $F(s) = (\log \Xi^{\mathrm{fin}})'(s) = \frac{1}{2}(B^{\mathrm{fin}}(s) - B^{\mathrm{fin}}(1-s)) + (\log \mathbf{G}_H)'(s)$. Since $\mathbf{G}_H(1-s) = \mathbf{G}_H(s)$, $(\log \mathbf{G}_H)'(1-s) = -(\log \mathbf{G}_H)'(s)$. Also $B^{\mathrm{fin}}(1-(1-s)) = B^{\mathrm{fin}}(s)$, so $B^{\mathrm{fin}}(s) - B^{\mathrm{fin}}(1-s)$ is odd. Hence $F(1-s) = -F(s)$ and integrating from $1/2$ gives $\Xi^{\mathrm{fin}}(1-s) = \Xi^{\mathrm{fin}}(s)$.

### Explicit formula (residues and band)

Consider

$$I = \frac{1}{2\pi i} \int_{(1+\varepsilon)} \frac{\Xi^{\mathrm{fin}\prime}(s)}{\Xi^{\mathrm{fin}}(s)} \frac{x^s}{s} \widehat{K}_H\Big(\frac{s-\frac{1}{2}}{i}\Big)\, ds.$$

Shift to $\Re s = -\varepsilon$; residues at $s = \rho$ (zeros) and $s = 0$ contribute. Since $\widehat{K}_H \geq 0$ and compactly supported in frequency, only zeros with $|\Im(\rho - \frac{1}{2})| \leq 2c/H$ appear and each contributes $-\frac{x^\rho}{\rho}\widehat{K}_H((\rho - \frac{1}{2})/i)$. Identifying the prime side via $\Xi^{\mathrm{fin}\prime}/\Xi^{\mathrm{fin}}$ and reversing the Mellin transform yields Theorem 7.1.

# B Appendix B: JSON Certificate Schemas

**Bundle fields (illustrative).**

```
{
  "X": "float", "A": "float", "gamma": "float", "c": "float", "B": "float",
  "H": "float", "Q": "float", "N": "integer",
  "kernel": {"JH": {"type": "gaussian", "alpha": "1/H"},
             "KH": {"type": "fejer", "band": "2*c/H"}},
  "bank": {"M": "integer", "mask_v": "array, sum v = sum j v = 0, ||v||2=1"},
  "quadrature": {"xi": "array", "w": "array", "t": "array", "h": "float", "delta": "float"},
  "eta": "float or symbol"
}
```

**Certificates (illustrative).**

```
{
  "fe_cert": {"max_diff": "float", "tolerance": "float", "passed": "bool"},
  "psd_cert": {"lambda_min": "float", "norm2": "float", "tau_psd": "float", "passed": "bool"},
  "margin_cert": {
    "c0": "float", "epsH": "float", "Hinv": "float", "e_minus_B": "float",
    "C1": "float", "C2": "float", "C3": "float", "M_H": "float",
    "tau_sum": "float", "passed": "bool"
  },
  "frame_cert": {"h": "float", "delta": "float", "band": "float",
                 "C_frame": "float", "passed": "bool"}
}
```

## C Appendix C: Implementation Notes

**Determinism.** Fix summation orders and use pairwise summation trees or Kahan summation. Record hashes of inputs/outputs.

**PSD numerics.** Certify PSD via $\lambda_{\min} \geq \tau_{\mathrm{PSD}}$; report $\|\mathbf{K}\|_2$ and matrix size.

**Sampling.** Use near-uniform $\xi_j$ with step $h \leq \pi H/4$ and jitter $\delta < 1/4$; choose $t_k$ with mesh $\ll 1/H$ on the same band.

## A Completed Proofs and Auxiliary Lemmas

### A.1 Finite functional equation for $\Xi_{\mathrm{fin}}$

Recall the definitions in §4:

$$B_H^{\mathrm{fin}}(s) = \sum_{2 \leq n \leq N} b_{H,Q}(n)\, n^{-s}, \qquad \Xi_{\mathrm{fin}}(s) \overset{\mathrm{def}}{=} \exp\Big(\frac{1}{2}\int_{1/2}^{s} \big(B_H^{\mathrm{fin}}(u) - B_H^{\mathrm{fin}}(1-u)\big)\, du\Big)\, G_H(s),$$

where $G_H$ is Mellin-even and nonvanishing: $G_H(1-s) = G_H(s) \neq 0$ for all $s$.

**Theorem A.1** (Exact FE)**.** $\Xi_{\mathrm{fin}}(1-s) = \Xi_{\mathrm{fin}}(s)$ *for all* $s \in$.

*Proof.* Differentiating $\log \Xi_{\mathrm{fin}}$ yields

$$\frac{(\Xi_{\mathrm{fin}})'}{\Xi_{\mathrm{fin}}}(s) = \frac{1}{2}\big(B_H^{\mathrm{fin}}(s) - B_H^{\mathrm{fin}}(1-s)\big) + \frac{G_H'}{G_H}(s).$$

By Mellin-evenness $\big(\log G_H\big)'(1-s) = -\big(\log G_H\big)'(s)$. Hence

$$\frac{d}{ds}\log\frac{\Xi_{\mathrm{fin}}(1-s)}{\Xi_{\mathrm{fin}}(s)} = -\frac{1}{2}\big(B_H^{\mathrm{fin}}(1-s) - B_H^{\mathrm{fin}}(s)\big) - \frac{G_H'}{G_H}(1-s) - \frac{1}{2}\big(B_H^{\mathrm{fin}}(s) - B_H^{\mathrm{fin}}(1-s)\big) - \frac{G_H'}{G_H}(s) = 0.$$

Thus $\Xi_{\mathrm{fin}}(1-s) = C\,\Xi_{\mathrm{fin}}(s)$ for a constant $C$. Evaluating at $s = \frac{1}{2}$ gives $C = 1$. □

### A.2 Smoothed explicit formula with sign and band-limited weights

Let $K_H$ be the Fejér log-kernel with $\widehat{K}_H = \widehat{K}_H \geq 0$ even and supported in $|t| \leq 2c/H$. Define

$$T_H(x) \;=\; \sum_{n \leq N} b_{H,Q}(n)\, K_H(\log n - \log x).$$

**Theorem A.2** (Smoothed EF; minus/weight structure)**.** *For* $x > 0$*,*

$$T_H(x) = \mathrm{Main}_H(x) \;-\; \sum_{\rho} \widehat{K}_H\Big(\frac{\rho - \frac{1}{2}}{i}\Big)\, \Re\Big(\frac{x^{\rho}}{\rho}\Big) \;+\; \mathcal{G}_H(x) \;+\; \mathcal{E}_H^{\mathrm{fin}}(x),$$

*where the sum runs over zeros* $\rho$ *of* $\Xi_{\mathrm{fin}}$*, the multiplier* $\widehat{K}_H \geq 0$ *is compactly supported,* $\mathcal{G}_H$ *is the completion term from* $G_H$ *and* $\mathcal{E}_H^{\mathrm{fin}}$ *is a finite-cutoff error (absorbed into the ledger). The zero sum enters with a global minus sign and nonnegative weights.*

*Proof.* Convolve $(\log \Xi_{\rm fin})'(s)$ against $\frac{x^s}{s}\widehat{K}_H((s-\frac{1}{2})/i)$ on $\Re s = \sigma$ and use Cauchy's theorem to shift across the zeros of $\Xi_{\rm fin}$. The odd part $\frac{1}{2}(B_H^{\rm fin}(s) - B_H^{\rm fin}(1-s))$ produces $T_H(x)$ after Mellin inversion; the poles/zeros contribute $-\sum \widehat{K}_H((\rho-\frac{1}{2})/i)\Re(x^\rho/\rho)$ because $\widehat{K}_H$ is even and real; the $G_H$-piece defines $\mathcal{G}_H(x)$. Truncation at $N$ yields $\mathcal{E}_H^{\rm fin}$. The compact $t$-support and $\widehat{K}_H \geq 0$ certify nonnegativity of weights and band-limitation. □

## A.3 Transfer from $J_H$ to $K_H$ in $L^2(dx/x)$

**Lemma A.1** (Transfer bound)**.** *Let $T_H$ and $\Psi_{H,Q}$ be the $K_H$– and $J_H$–smoothed functionals. Then on any dyadic $[X, 2X]$,*

$$\|T_H - \Psi_{H,Q}\|_{L^2([X,2X],dx/x)} \ll H^{-1}.$$

*Proof.* Identical to the proof provided earlier (Appendix A.2 uses Plancherel on the log-line): $\|\widehat{K_H} - \widehat{J_H}\|_{L^\infty} \ll H^{-1}$ and the $t$–integration runs over length $O(1/H)$. □

## A.4 Carrier-shift invariance (height scanning)

**Proposition A.1.** *Set $K_{H,\tau}(z) = e^{i\tau z}K_H(z)$, $\widehat{K_{H,\tau}}(t) = \widehat{K}_H(t-\tau)$. Then Theorem A.2 holds with $K_H \mapsto K_{H,\tau}$ and the zero-weights $\widehat{K}_H((\rho-\frac{1}{2})/i)$ replaced by $\widehat{K}_H(\Im\rho - \tau)$. The band is the translate $\{|\xi-\tau| \leq 2c/H\}$.*

*Proof.* Multiplication by $e^{i\tau z}$ shifts frequency by $\tau$, preserving evenness and nonnegativity. Phases $x^{-i\tau}$ factor from the main/gamma pieces; norms and signs are unchanged. □

## A.5 Frame inequality for Fejér atoms (discrete Paley–Wiener sampling)

Let $B_H = [-2c/H, 2c/H]$, $\Phi_k(\xi) = \widehat{K}_H(\xi - t_k)$ with step $h \leq \pi H/4$ and jitter $\delta < 1/4$.

**Lemma A.2** (Frame lower bound)**.** *There exists $C_{\rm frame} > 0$ depending only on $(h, \delta, \widehat{K}_H)$ such that for all finitely supported $a = (a_k)$,*

$$\int_{B_H} \Big| \sum_k a_k \Phi_k(\xi) \Big|^2 d\xi \ \geq\ C_{\rm frame} \sum_k |a_k|^2.$$

*Proof sketch (standard).* $\Phi_k$ are shifts of a compactly supported generator in a shift-invariant space on $B_H$. Under Nyquist-safe spacing and jitter $< 1/4$, Kadec's 1/4-theorem (and its compactly supported refinements) yield stability of sampling; the Riesz (frame) lower bound is continuous in $(h, \delta)$. Restricting to $B_H$ and using $\widehat{K}_H \geq 0$ preserves the bound. (A complete proof may be given by periodization on a torus of length $\asymp 1/H$ and diagonalizing the Gramian.) □

## A.6 PSD kernel and coercivity

Define $\mathbf{B}_{jk} = (2\pi)^{-1/2}\,\Xi_{\rm fin}(\frac{1}{2} + i\xi_j)\,\Phi_k(\xi_j)$, $\mathbf{W} = \mathrm{diag}(w_j)$, and $\mathbf{K} = \mathbf{B}^*\mathbf{W}\mathbf{B}$.

**Proposition A.2** (PSD & coercivity)**.** $\mathbf{K}$ *is PSD. If $|\Xi_{\rm fin}(\frac{1}{2}+i\xi)| \geq m_H > 0$ on $B_H$, then*

$$a^*\mathbf{K}a \ \geq\ \frac{m_H^2}{2\pi}\, C_{\rm frame}\, \|a\|_2^2$$

*for all $a \in^M$, where $C_{\rm frame}$ is from Lemma A.2.*

*Proof.* PSD: Gram structure. Coercivity: continuous inequality follows from $|F| \geq m_H$ and the frame bound; quadrature stability transfers it to the discrete matrix. □

### A.7 Alignment on a dyadic

**Lemma A.3** (Phase alignment)**.** *For $\rho = \beta + i\gamma$ and $x \in [X, 2X]$,*

$$\max_{x\in[X,2X]} \Re\Big(\frac{x^\rho}{\rho}\Big) \;\geq\; \cos\Big(\frac{|\gamma|\,\log 2}{1}\Big)\,\frac{X^\beta}{|\rho|},$$

*and if $|\gamma| \leq 2c/H$, $\cos(|\gamma|\log 2) = 1 - O(H^{-2})$.*

*Proof.* $\Re(x^\rho/\rho) = x^\beta|\rho|^{-1}\cos(\gamma\log x - \arg\rho)$. As $\log x$ spans an interval of length $\log 2$, the cosine attains $\geq \cos(|\gamma|\log 2)$. A Taylor expansion at 0 gives $1 - O((|\gamma|\log 2)^2) = 1 - O(H^{-2})$ in-band. □

### A.8 Finite positivity barrier (full proof)

Let $M_H = c_0 - C_1\varepsilon_H - C_2H^{-1} - C_3e^{-B}$ and $C_\Gamma = \sup_x |\mathcal{G}_H(x)|$.

**Theorem A.3** (Zero-free band under finite margin)**.** *If $M_H > C_\Gamma + x^{1/2}\,\mathsf{S}_H^{\mathrm{cert}}$ for some $x \in [X, 2X]$ where*

$$\mathsf{S}_H := \sum_\rho \widehat{K}_H\Big(\frac{\rho - \frac{1}{2}}{i}\Big)\frac{1}{|\rho|} \quad \textit{and} \quad \mathsf{S}_H^{\mathrm{cert}} \geq \mathsf{S}_H,$$

*then $\Xi_{\mathrm{fin}}$ has no zero with $\Re s \neq \frac{1}{2}$ and $|\Im(s - \frac{1}{2})| \leq 2c/H$.*

*Proof.* Assume a zero $\rho_0 = \beta_0 + i\gamma_0$ with $\beta_0 \neq 1/2$ in band. By Lemma A.3 choose $x$ with $\Re(x^{\rho_0}/\rho_0) \geq \kappa_{\mathrm{al}}x^{1/2}/|\rho_0|$, $\kappa_{\mathrm{al}} = 1 - O(H^{-2})$. Take real parts in Theorem A.2 and rearrange:

$$\sum_\rho \widehat{K}_H\Big(\frac{\rho - \frac{1}{2}}{i}\Big)\Re\Big(\frac{x^\rho}{\rho}\Big) \;\leq\; \mathrm{Main}_H(x) - T_H(x) + C_\Gamma.$$

By calibration $\mathrm{Main}_H(x) \leq c_0 + O(H^{-1})$ and by the ledger $T_H(x) \geq c_0 - (C_1\varepsilon_H + C_2H^{-1} + C_3e^{-B}) = M_H$. Hence

$$\sum_\rho \widehat{K}_H\Big(\frac{\rho - \frac{1}{2}}{i}\Big)\Re\Big(\frac{x^\rho}{\rho}\Big) \;\leq\; C_\Gamma - M_H.$$

Split off $\rho_0$ and bound the rest by $\leq x^{1/2}\mathsf{S}_H$:

$$\kappa_{\mathrm{al}}\,\widehat{K}_H\Big(\frac{\rho_0 - \frac{1}{2}}{i}\Big)\frac{x^{1/2}}{|\rho_0|} \;\leq\; C_\Gamma - M_H + x^{1/2}\mathsf{S}_H \;\leq\; C_\Gamma - M_H + x^{1/2}\mathsf{S}_H^{\mathrm{cert}}.$$

Since the left side is nonnegative and $\kappa_{\mathrm{al}} \leq 1$, this contradicts $M_H > C_\Gamma + x^{1/2}\mathsf{S}_H^{\mathrm{cert}}$. □

### A.9 Completion term bound

**Proposition A.3.** *For the Mellin-even Gaussian completion $G_H$ and Fejér $K_H$ with compact $t$-support,*

$$C_\Gamma := \sup_{x>0}|\mathcal{G}_H(x)| \;\ll\; 1 + \frac{c^2}{H^2}.$$

*Proof.* As in Proposition 15.6: truncate $t$ to $O(1/H)$; $(\log G_H)'$ is affine in $t$, giving $O(1 + c^2/H^2)$ uniformly in $x$. □

## A.10 Finite-to-classical comparison on $\Re s > 1$

Let $\psi \in C_c^\infty()$ with $\|\psi\|_{H^1}, \|\psi\|_{L^2} \le 1$.

**Proposition A.4** (Test-function agreement). *Uniformly on $\sigma > 1$,*

$$\int \Big( B_H^{\rm fin} - \frac{\zeta'}{\zeta} \Big)(\sigma + it)\, \psi(t)\, dt \;=\; O\Big( H^{-1} + \varepsilon_H + e^{-B} \Big).$$

*Proof.* Split $B_H^{\rm fin} = \sum_{n \le N} b_{H,Q}(n)\, n^{-s}$. The truncation tail for $-\zeta'/\zeta$ beyond $N$ is $O(e^{-B})$; the $J_H \to K_H$ replacement contributes $O(H^{-1})$ (Lemma A.1); ETI variance gives $O(\varepsilon_H)$ after testing against $\psi$; the $H^1$ bound ensures all integrations by parts are controlled. □

## A.11 Goldbach-only mode is sufficient for all analytic steps

**Proposition A.5.** *All statements in §7, §8, §9, §10 hold when $b_{H,Q}$ is replaced by $b_{H,Q}^{\rm GB}$ and ETI$(X; H, Q)$ is replaced by ETI$_{GB}(X; H, Q)$.*

*Proof.* The proofs use only the pin/ledger/AO–SSU/short-shift bounds and never invoke a single-shift/twin input. The constructions of $B_H^{\rm fin}$, $\Xi_{\rm fin}$, the EF, the PSD kernel and the margin/barrier are algebraic in the finite coefficients and transfer verbatim. □

## A.12 Non-coercive envelope fallback (argument principle)

**Proposition A.6.** *Let $\mathcal{R}_H(\tau) = \{\sigma + i\xi : \frac{1}{2} - \sigma_0 \le \sigma \le \frac{1}{2} + \sigma_0,\ |\xi - \tau| \le 2c/H\}$ and*

$$N_H(\tau) = \frac{1}{2\pi i} \oint_{\partial \mathcal{R}_H(\tau)} \frac{\Xi'_{\rm fin}(s)}{\Xi_{\rm fin}(s)}\, ds.$$

*Then a certified envelope bound is*

$$\mathsf{S}_{H,\tau}^{\rm cert} \;\le\; \sum_{k=1}^{N_H(\tau)} \frac{1}{|\rho_k|} \;\le\; \frac{2 N_H(\tau)}{1} \;=\; 2\, N_H(\tau),$$

*where the $\rho_k$ are zeros in $\mathcal{R}_H(\tau)$. This bound is independent of any $m_H$ assumption.*

*Proof.* $|\rho| \ge 1/2$ on $\mathcal{R}_H(\tau)$; since $0 \le \widehat{K}_H \le 1$, $\mathsf{S}_{H,\tau} \le \sum 1/|\rho_k|$; counting $N_H(\tau)$ by the argument principle yields the claim. □

## A.13 Ledger decomposition and deterministic constants

**Lemma A.4** (Ledger decomposition). *With $H = (\log X)^A$, $Q = H^\gamma$, $N = 2Xe^{B/H}$,*

$$\mathcal{L}_H \;=\; C_1\, \varepsilon_H \;+\; C_2\, H^{-1} \;+\; C_3\, e^{-B}, \qquad \varepsilon_H \asymp \Big(\frac{H}{X}\Big)^{1/2} + \frac{1}{HQ^2}.$$

*Here $C_2, C_3$ depend only on the fixed kernel family and the schedule; $C_1$ multiplies the ETI scale.*

*Proof.* $C_2 H^{-1}$ accounts for $\widehat{K_H} - \widehat{J_H}$ and band tails (Appendix A.3); $C_3 e^{-B}$ is the Dirichlet tail beyond $N$; $C_1 \varepsilon_H$ is the ETI variance on $J_H$-smoothing transferred to $K_H$. □

### A.14 Uniformity in the carrier

**Lemma A.5** (Uniform-in-$\tau$ formulation)**.** *Suppose ETI$(X; H, Q)$ and the numerical envelope routine hold uniformly for $\tau$ in compact sets. Then Theorems A.2 and A.3 hold uniformly for all translated bands $|\xi - \tau| \le 2c/H$, yielding zero-free bands for $\Xi_{\mathrm{fin}}$ centered at any such $\tau$ where the margin inequality passes.*

*Proof.* By Proposition A.1 all inputs (band width, nonnegativity, frame, completion bound) are $\tau$-invariant. Uniform ETI and uniform envelope bounds give a uniform margin test. $\square$

## References